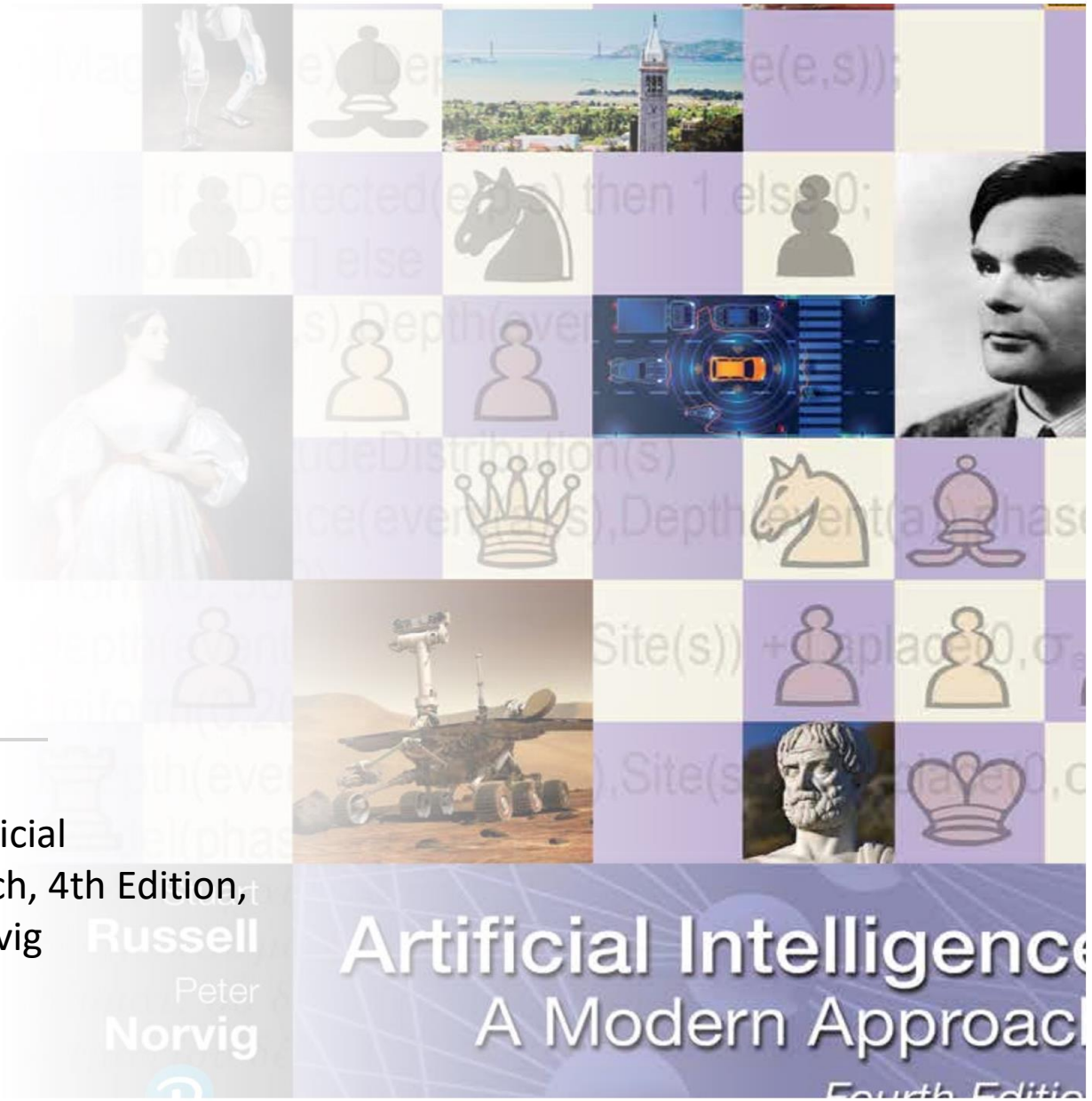
BCSE306L
Artificial Intelligence

Introduction



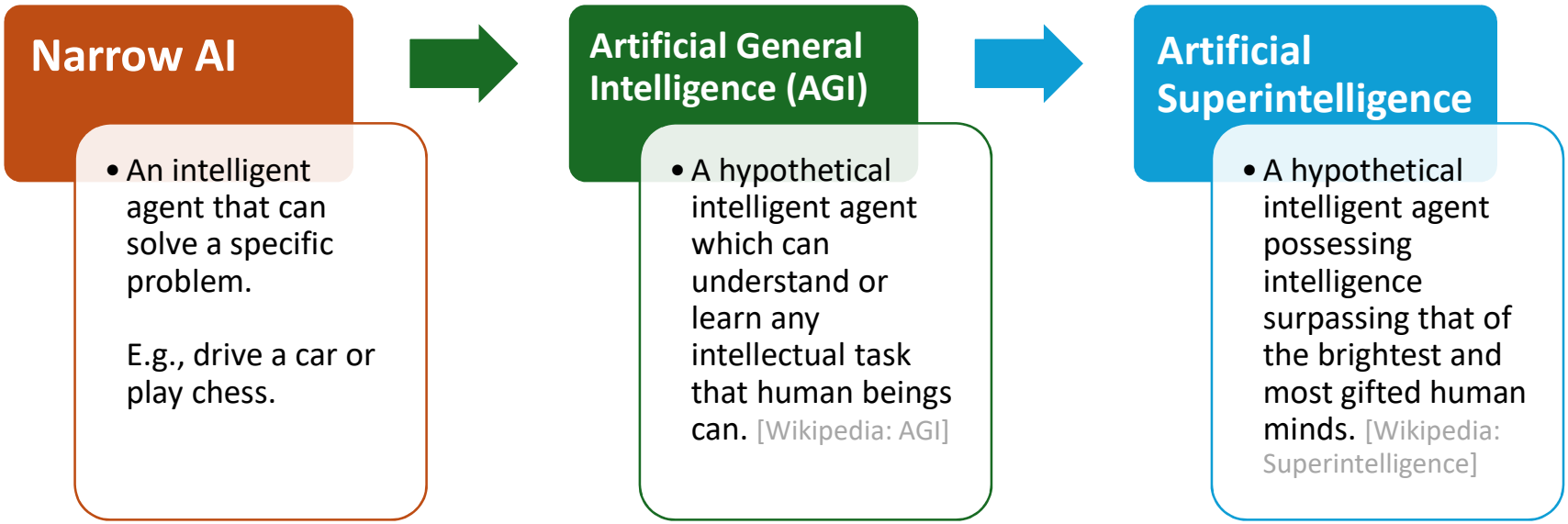
Primary Textbook: [**AIMA**] Artificial Intelligence: A Modern Approach, 4th Edition, by Stuart Russell and Peter Norvig

What is Artificial Intelligence?

- Artificial Intelligence is the ability of a machine or computer system to perform tasks that normally require human intelligence – **common definition**
- According to **Russell and Norvig (AIMA)**: **Artificial Intelligence is the study of the components required to build a rational (intelligent) agent.**
- An **intelligent (rational) agent** is: **An entity that perceives its environment through sensors and acts upon that environment through actuators to achieve its goals.**
- Artificial Intelligence is the study and design of rational agents that perceive their environment and take actions to maximize the achievement of their goals.

The Goal of AI

"Have machines solve problems that are challenging for humans."
We call such a machine an **intelligent agent.**



How can we achieve this? Create an agent that can:

Think like a human? | Act like a human? | Think rationally? | Act rationally?

Think like a human?

Act like a human?

Think rationally?

Act rationally?

The brain as an information processing machine.

- Requires scientific theories of **how the brain works**.

Note: The brain does not work like artificial neural networks from machine learning!

How to understand cognition as a computational process?

- Introspection: try to think about how we think.
- **Predict the behavior of human** subjects.
- Image the brain, examine neurological data

AI consciousness

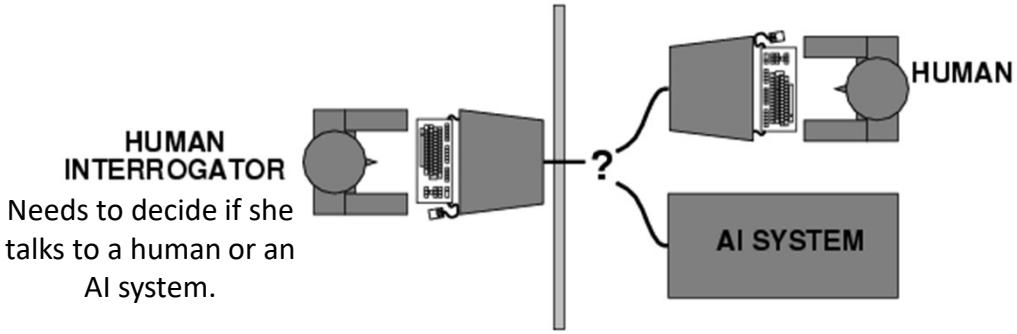
- What does it mean that a machine is conscient/sentient?
- How can we tell?
- What do we do then?

Cognitive Sciences

Moral Philosophy

Think like a human? | **Act like a human?** | Think rationally? | Act rationally?

- Alan Turing rejects the question “Can machines think?”
- The Turing Test tries to define what acting like a human means.





Alan Turing (1950) “Computing machinery and intelligence”

- What capabilities would a computer need to have to pass the Turing Test? These are still the core AI areas.
  - Natural language processing
  - Knowledge representation
  - Automated reasoning
  - Machine learning
- Turing predicted that by the year 2000, machines would be able to fool 30% of human judges for five minutes.
  ChatGPT (2023) is probably doing a least that!

Turing Test: Criticism

What are some potential problems with the Turing Test?

- Some human behavior is not intelligent.
- Some intelligent behavior may not be human.
- Human observers may be easy to fool.
  - A lot depends on expectations.
  - Anthropomorphic fallacy: humans tend to humanize things.
- Imitate intelligence without intelligence. E.g., the early chatbots ELIZA (1964) simulates a conversation using pattern matching.

Is passing the Turing test a good scientific goal?

- Engineering perspective: Imitating a human is not a good way to solve practical problems.
- We can create useful intelligent agents without trying to imitate humans.

Chinese Room Argument



Thought experiment by John Searle (1980): Imitate intelligence using rules.

Think like a human? | Act like a human? | **Think rationally?** | Act rationally?

- **Thinking Rationality**: Draw sensible conclusions from facts, logic and data.
- **Logic**: A chain of argument that always yield correct conclusions.
  E.g., "Socrates is a man; all men are mortal; therefore, Socrates is mortal."
- **Logic-based approach to AI**: Describe a problem in formal logic notation and apply general deduction procedures to solve it.
  Issues:
  - Describing real-world problems and knowledge using logic notation is hard.
  - Computational complexity of finding the solution.
  - Much intelligent or "rational" behavior in an uncertain world cannot be defined by simple logic rules.

Example: What about the logical implication

$$study\ hard \Rightarrow A\ in\ my\ AI\ course$$

Should it rather be

$$study\ hard\ AND\ be\ lucky\ AND\ \ldots \Rightarrow A\ in\ my\ AI\ course$$

Think like a human?

Act like a human?

Think rationally?

Act rationally?

Acting rational means to try to achieve the “best” outcome.

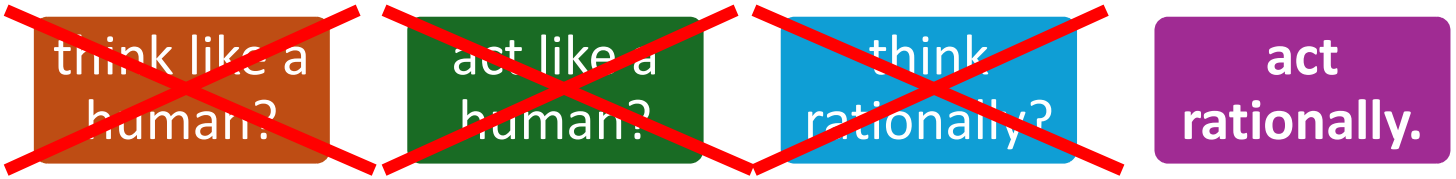
- Best means that we need to do **optimization**.
- The desirability of outcomes can be measured by the economic concept of **utility.** If there is uncertainty about achieving outcomes, then we need to maximizing the **expected utility.**

- Optimization has several advantages:
  - **Generality**: optimization is not limited to logical rules.
  - **Practicality**: can be adapted to many real-world problems.
  - **Well established**: existing solvers and methods for simulation and experimentation.
  - Avoids philosophy and psychology in favor of a **clearly defined objective**.

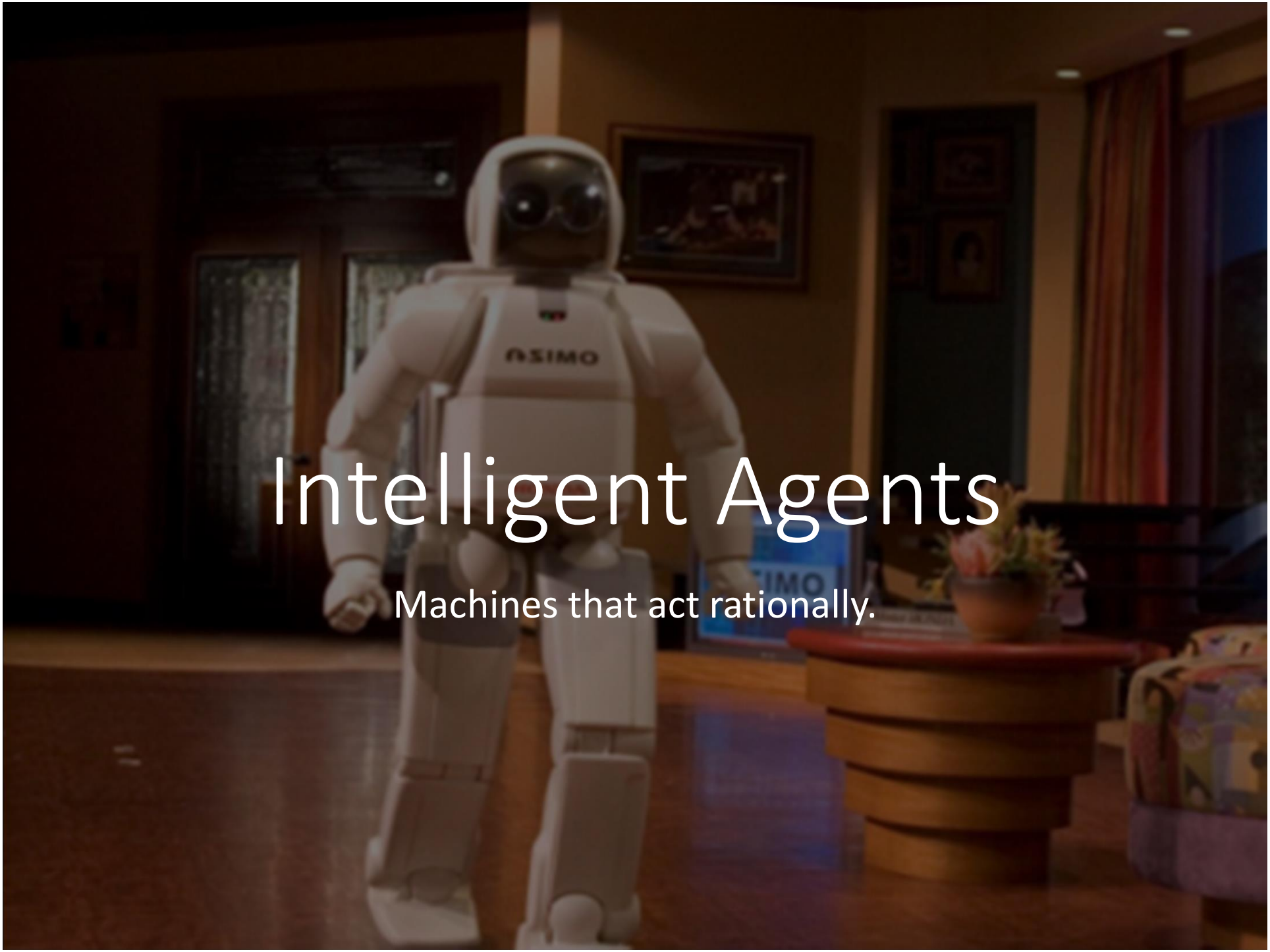
- **Bounded rationality:** In practice, expected utility optimization is subject to the agent’s knowledge and computational constraints. The agent needs to do the best with its knowledge and resources.

The Type of AI is Covered in this Course

This course focuses on the computational techniques needed to creating a **narrow AI agent** that can



That is, create a machines that acts in a way to solve a specific hard problem that traditionally would have been thought to require human intelligence.



Intelligent Agents

Machines that act rationally.

Components of an Intelligent Agent

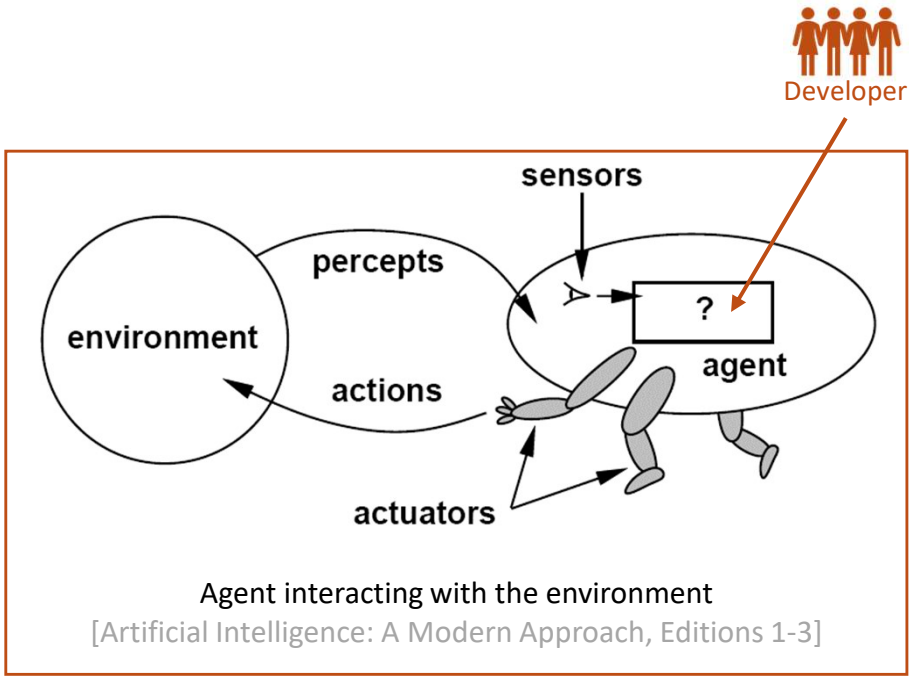
Intelligent agents **act rationally** in their environment.

They need to

- **Communicate** with the environment using **percepts** and **actions**.
- **Represent knowledge**, **reason** and **plan** to achieve a desired **outcome**.

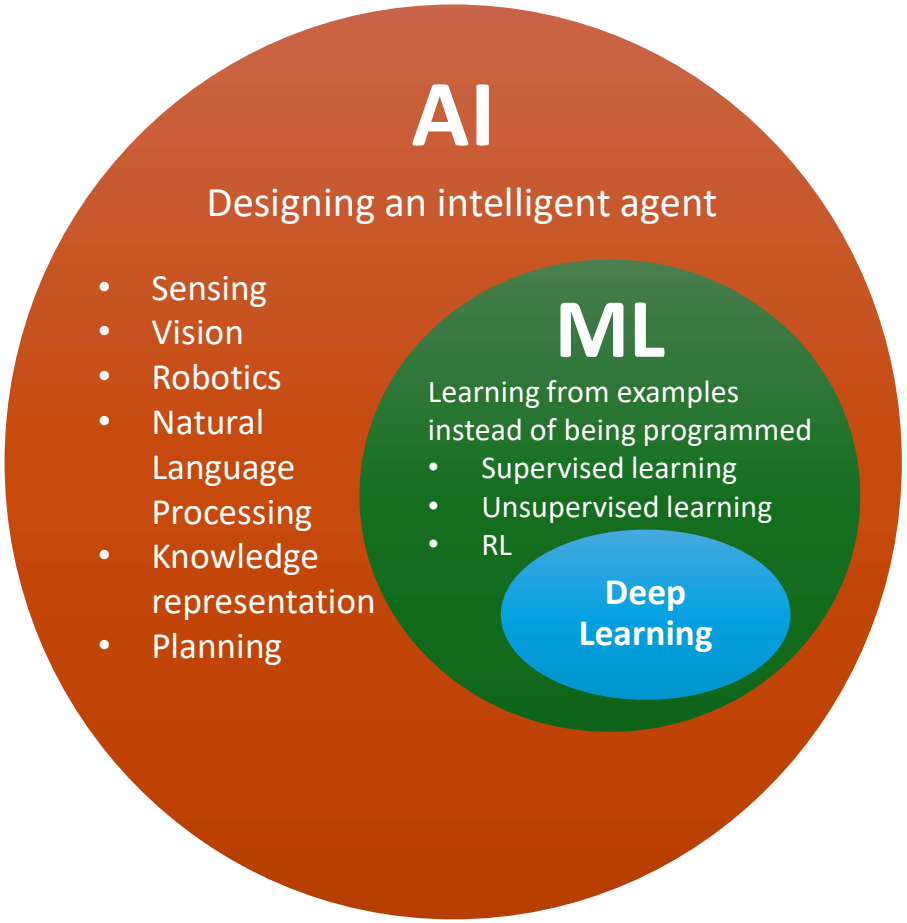
Optional

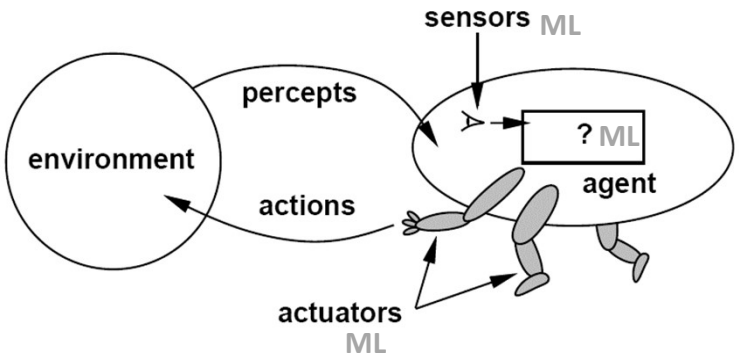
- **Learn from experience** to improve performance**.**



Agent interacting with the environment
[Artificial Intelligence: A Modern Approach, Editions 1-3]

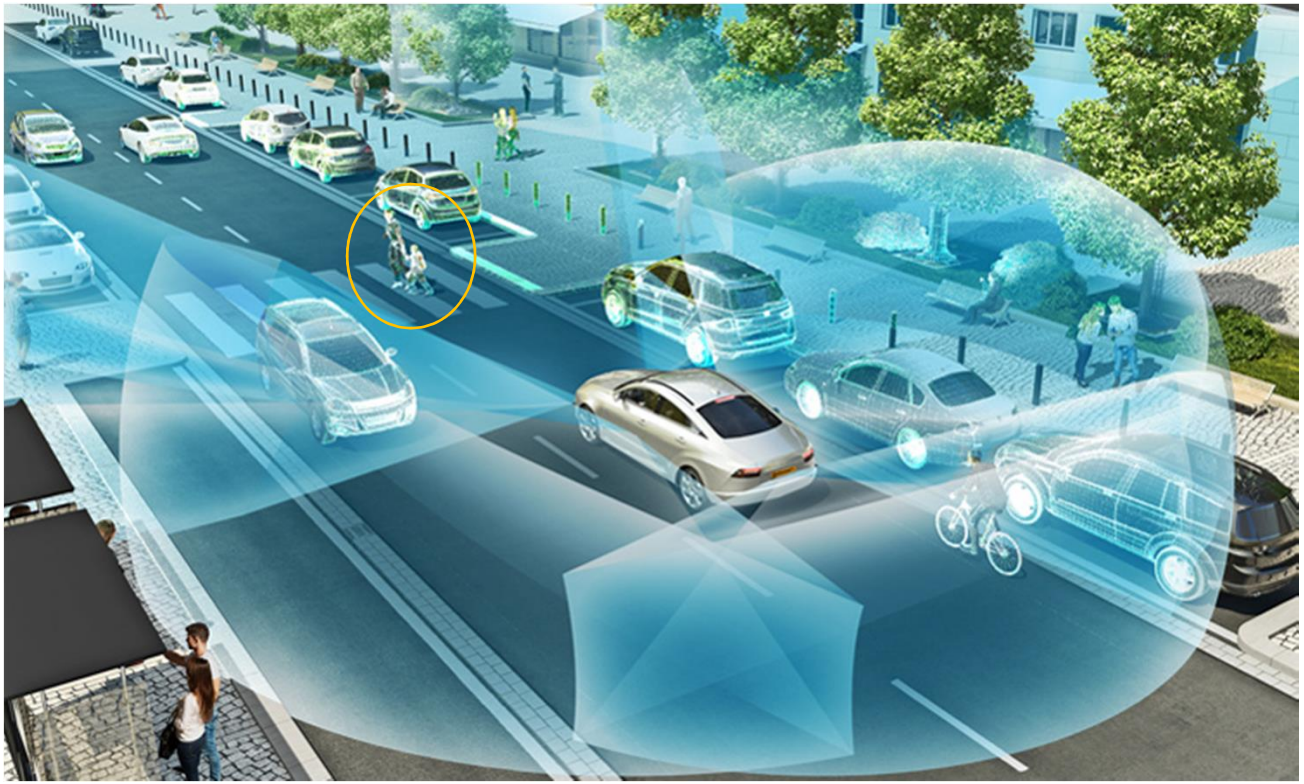
Artificial Intelligence vs. Machine Learning





Agent interacting with the environment
[Artificial Intelligence: A Modern Approach, Editions 1-3]

Example: Self-Driving Car

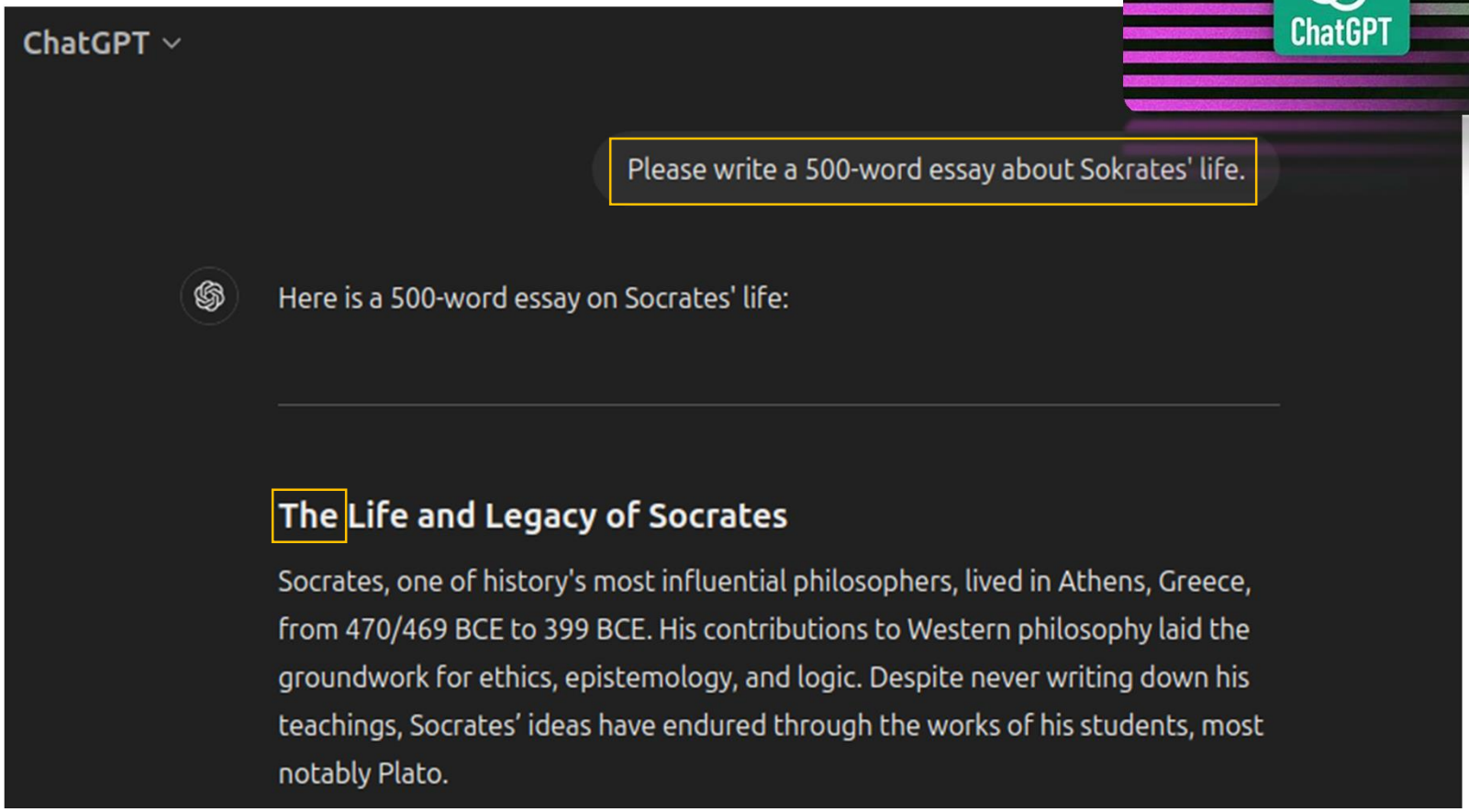


Percepts: Other objects including people crossing the street.

Action: Slow down or stop the car.

Objective: Reach the destination safely

Example: Homework and LLMs



Percept: Your prompt

Action: Next most likely word… More words are created word-by-word.

Objective: You may like a useful answer, but what is ChatGPT's objective?

How Each Topic of our Course Helps the Agent?

Example 1 : Imagine we're building a robot delivery agent that delivers packages.

- A **Robot Delivery Agent** is an **intelligent autonomous robot** that can **pick up, transport, and deliver packages** from one location to another without continuous human assistance. It uses Artificial Intelligence (AI) to **perceive its surroundings, make decisions, plan routes, and perform physical actions** safely and efficiently.

How a Robot Delivery Agent Works

- **Receives a delivery request** (e.g., deliver a package to Room 301).
- **Identifies the destination** using stored maps and building information.
- **Plans the best route** to reach the destination.
- **Navigates safely** by avoiding obstacles and people.
- **Delivers the package** to the correct location.
- **Returns** to its charging station or waits for the next task.

AI Component	Key Question	Simple Explanation	Example	Contribution
1. Search	How can the robot reach the destination?	Finds the best path among many possible routes.	Chooses the shortest route to Room 301.	Finds the best solution to achieve the goal.
2. Knowledge Representation	How does the robot remember things?	Stores information about the environment and rules.	Stores maps, room locations, and traffic rules.	Gives the robot knowledge and memory.
3. Logical Reasoning	How does the robot use its knowledge?	Uses stored knowledge to draw conclusions.	Red sign → Emergency exit.	Helps the robot make intelligent decisions.
4. Planning	How should the robot achieve its goal?	Creates a sequence of actions before acting.	Pick package → Go to elevator → Deliver package.	Organizes actions to complete the task.
5. Probability	What if the robot is uncertain?	Estimates the most likely situation using probabilities.	Detects an object as 80% likely to be a person.	Helps make decisions under uncertainty.

AI Component	Key Question	Simple Explanation	Example	Contribution
6. Decision Making (Utility)	Which action is best?	Chooses the action with the highest benefit.	Selects the safer route instead of the faster one.	Chooses the best action.
7. Machine Learning	Can the robot improve over time?	Learns from past experiences and improves performance.	Learns better routes after many deliveries.	Improves performance through learning.
8. Natural Language Processing (NLP)	How does the robot understand people?	Understands and responds to human language.	"Deliver this package to Room 301."	Enables communication with humans.
9. Computer Vision	How does the robot see?	Uses cameras to recognize objects and surroundings.	Detects doors, people, and obstacles.	Provides visual understanding of the environment.
10. Robotics	How does the robot perform actions?	Controls the robot's physical movements.	Moves, turns, picks up, and delivers packages.	Enables physical interaction with the real world.

Example 2:
Consider a self-driving car as an intelligent agent.

AI Subfield	Role in a Self-Driving Car
Perception (Computer Vision)	Detects roads, vehicles, pedestrians, traffic lights, lane markings, and other objects in the environment.
Knowledge Representation	Stores maps, traffic rules, road layouts, speed limits, and other driving-related knowledge.
Reasoning	Infers the appropriate action based on the situation, such as recognizing that a red traffic light means the vehicle must stop.
Search & Planning	Finds the best route from the source to the destination and plans safe paths while avoiding obstacles and traffic.
Probability	Estimates uncertainties, such as the likelihood that a pedestrian may cross the road or another vehicle may change lanes.
Utility-Based Decision Making	Chooses actions that maximize safety, comfort, and efficiency while minimizing travel time and risk.
Learning	Improves driving performance by learning from past experiences, driving data, and changing road conditions.
Robotics	Controls the physical actions of the vehicle, including steering, acceleration, braking, and vehicle stability.

Another Example 3: Chess Agent

Capability	Role
Search	Explores possible moves.
Knowledge Representation	Represents board positions.
Reasoning	Evaluates tactical situations.
Learning	Learns better strategies from games.
Decision Making	Selects the move with the highest chance of winning.

An intelligent agent perceives its environment and takes actions to achieve its goals. To do this effectively, the agent needs several abilities (also called as sub-fields of AI):

Topic/Chapter	How it helps an Intelligent Agent
Search	Finds a sequence of actions to reach a goal.
Knowledge Representation	Stores information about the world.
Logical Reasoning	Draws conclusions from known facts.
Planning	Decides future actions to achieve goals.
Uncertainty & Probability	Makes decisions when information is incomplete.
Decision Theory / Utility	Chooses the best action among alternatives.
Machine Learning	Improves performance through experience.
Natural Language Processing	Communicates with humans.
Computer Vision	Understands visual information from the environment.
Robotics	Acts physically in the environment.

- **Why AIMA (our course) Covers All These Topics**
- Russell and Norvig's main idea is:
  - **Intelligence is not one thing.**
  - An intelligent agent needs multiple abilities:
  - **Perceive → Represent Knowledge → Reason → Plan → Learn → Decide → Act**
  - Each major chapter of the book teaches one piece of this pipeline.
  - That's why the entire textbook can be viewed as:
    - **A study of the components/abilities required to build a rational (intelligent) agent.**

Applications of Artificial Intelligence (AI)

Artificial Intelligence is applied in almost every industry to automate tasks, improve decision-making, increase efficiency, reduce human effort, and solve complex real-world problems.

Application Area	How AI is Used	Example
Healthcare	Diagnoses diseases, analyzes medical images, and assists doctors.	Detecting cancer from X-rays or MRI scans.
Education	Provides personalized learning and intelligent tutoring.	AI-based learning platforms and virtual tutors.
Banking & Finance	Detects fraud, approves loans, and predicts market trends.	Credit card fraud detection.
E-Commerce	Recommends products and improves customer experience.	Product recommendations on shopping websites.
Transportation	Enables autonomous driving and traffic management.	Self-driving cars and route optimization.
Manufacturing	Automates production and predicts machine failures.	Industrial robots in assembly lines.
Agriculture	Monitors crops, detects diseases, and automates farming.	Drone-based crop monitoring.
Cybersecurity	Detects cyber threats and prevents attacks.	AI-based intrusion detection systems.
Entertainment	Recommends movies, music, and games.	Movie recommendations on streaming platforms.

Natural Language Processing (NLP)	Understands and generates human language.	Chatbots and virtual assistants.
Computer Vision	Recognizes faces, objects, and handwritten text.	Face recognition for attendance systems.
Robotics	Performs repetitive or dangerous physical tasks.	Warehouse and delivery robots.
Smart Homes	Automates home appliances using AI.	Smart speakers controlling lights and fans.
Retail	Predicts customer demand and manages inventory.	Automatic inventory management.
Weather Forecasting	Predicts weather conditions and natural disasters.	Rainfall and cyclone prediction.
Defense & Security	Performs surveillance and threat detection.	AI-powered surveillance systems.
Gaming	Controls intelligent game characters and adapts gameplay.	AI opponents in chess and video games.
Customer Service	Provides 24/7 automated customer support.	AI chatbots answering customer queries.
Logistics & Supply Chain	Optimizes delivery routes and warehouse operations.	AI-based package delivery planning.
Space Exploration	Assists in spacecraft navigation and planetary exploration.	AI-powered Mars rovers.

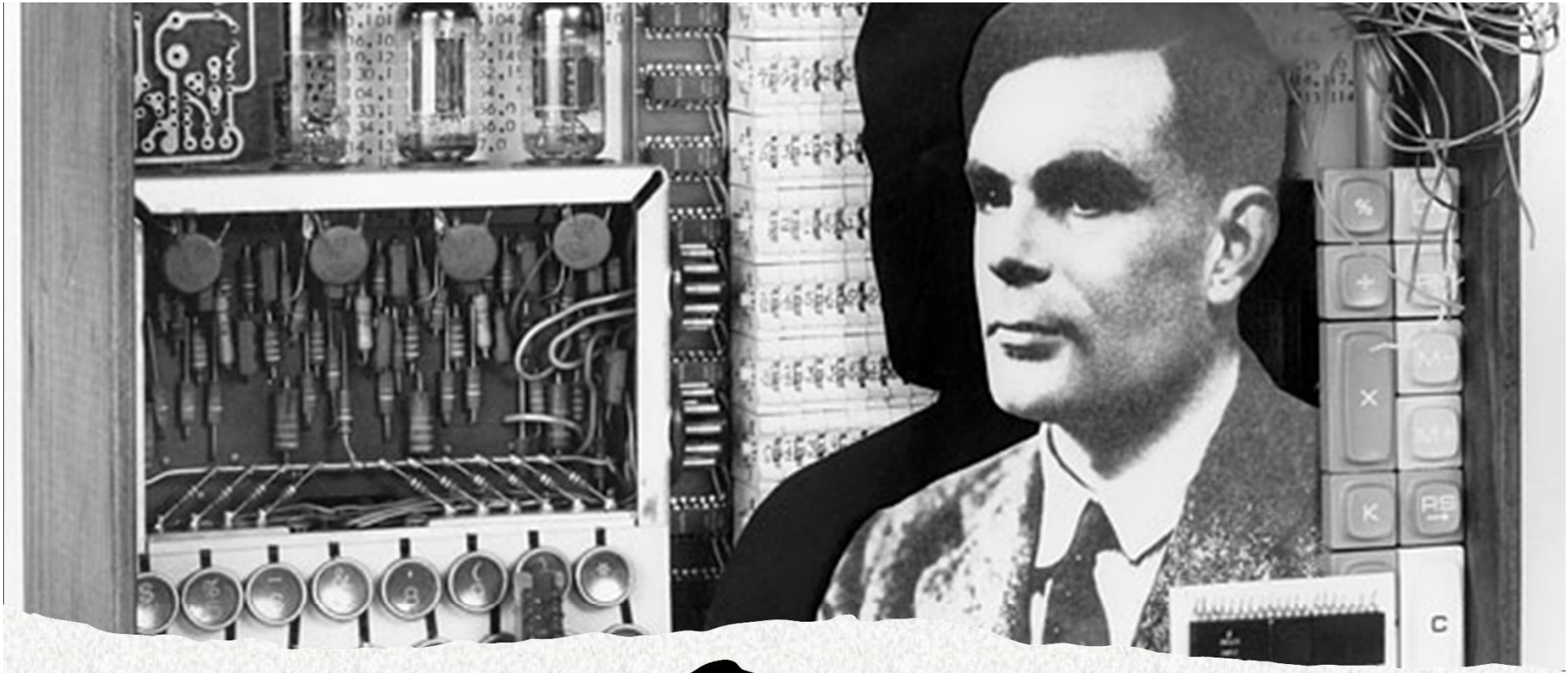
Types of Artificial Intelligence (AI)

Type of AI	Description	Capabilities	Examples
Narrow AI (Weak AI)	Designed to perform a **specific task**. It cannot perform tasks outside its programmed domain.	Solves one problem efficiently but cannot think or learn beyond its assigned task.	Chatbots, face recognition systems, voice assistants, recommendation systems, self-driving car features.
Artificial General Intelligence (AGI)	AI that can **understand, learn, reason, and perform any intellectual task** that a human can do.	Can adapt to new situations, solve different types of problems, and transfer knowledge across domains.	**Does not exist yet** (currently under research).
Artificial Super Intelligence (ASI)	AI that **surpasses human intelligence** in all areas, including creativity, reasoning, learning, and decision-making.	Can outperform humans in every intellectual task and may improve itself continuously.	**Hypothetical** (does not exist yet).

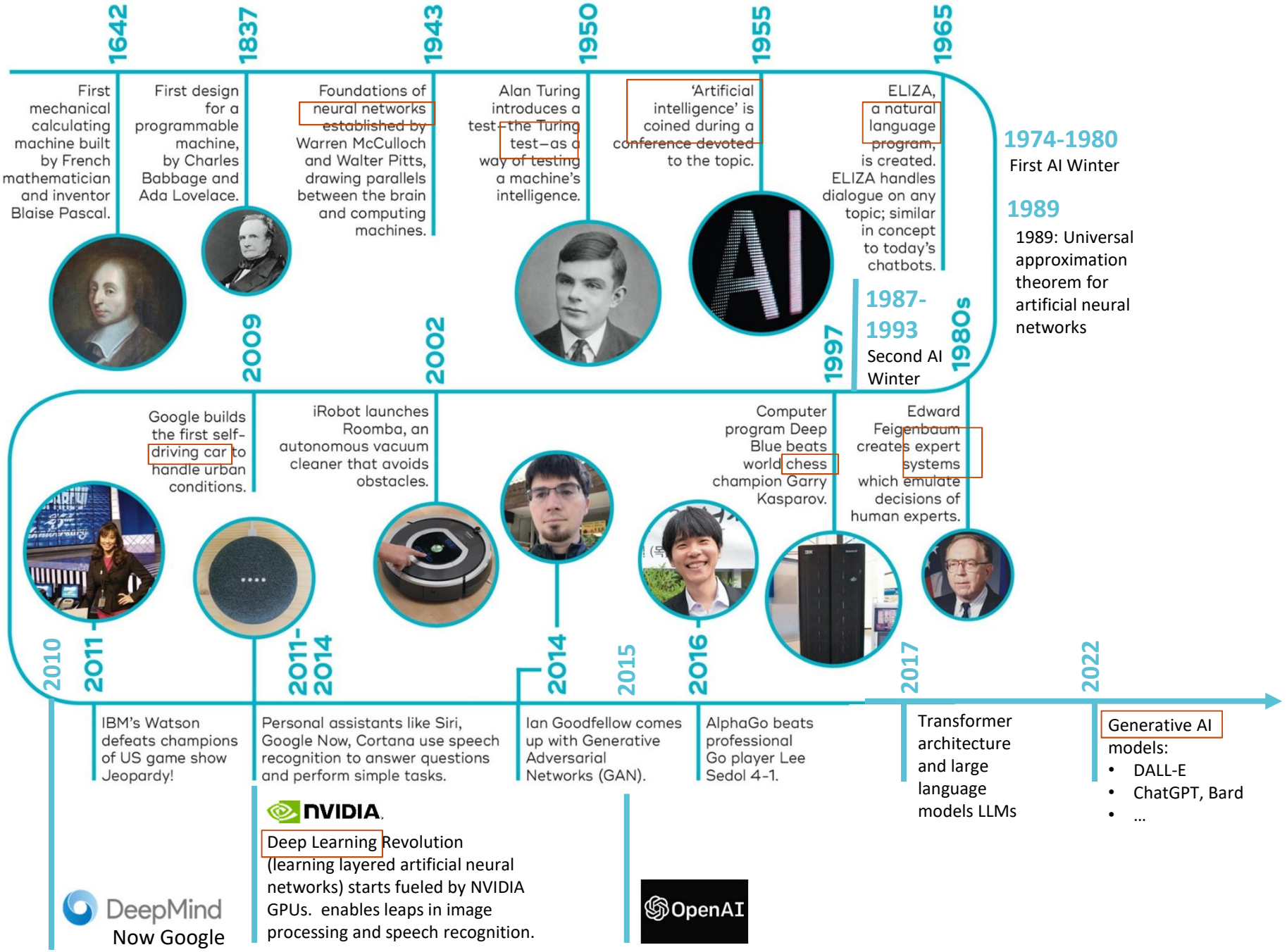
Key Differences

Feature	Narrow AI	AGI	ASI
Performs a specific task	✅ Yes	✅ Yes	✅ Yes
Performs multiple different tasks	❌ No	✅ Yes	✅ Yes
Human-level intelligence	❌ No	✅ Yes	✅ Exceeds humans
Learns and adapts like a human	Limited	✅ Yes	✅ Far beyond humans
Currently available	✅ Yes	❌ No	❌ No

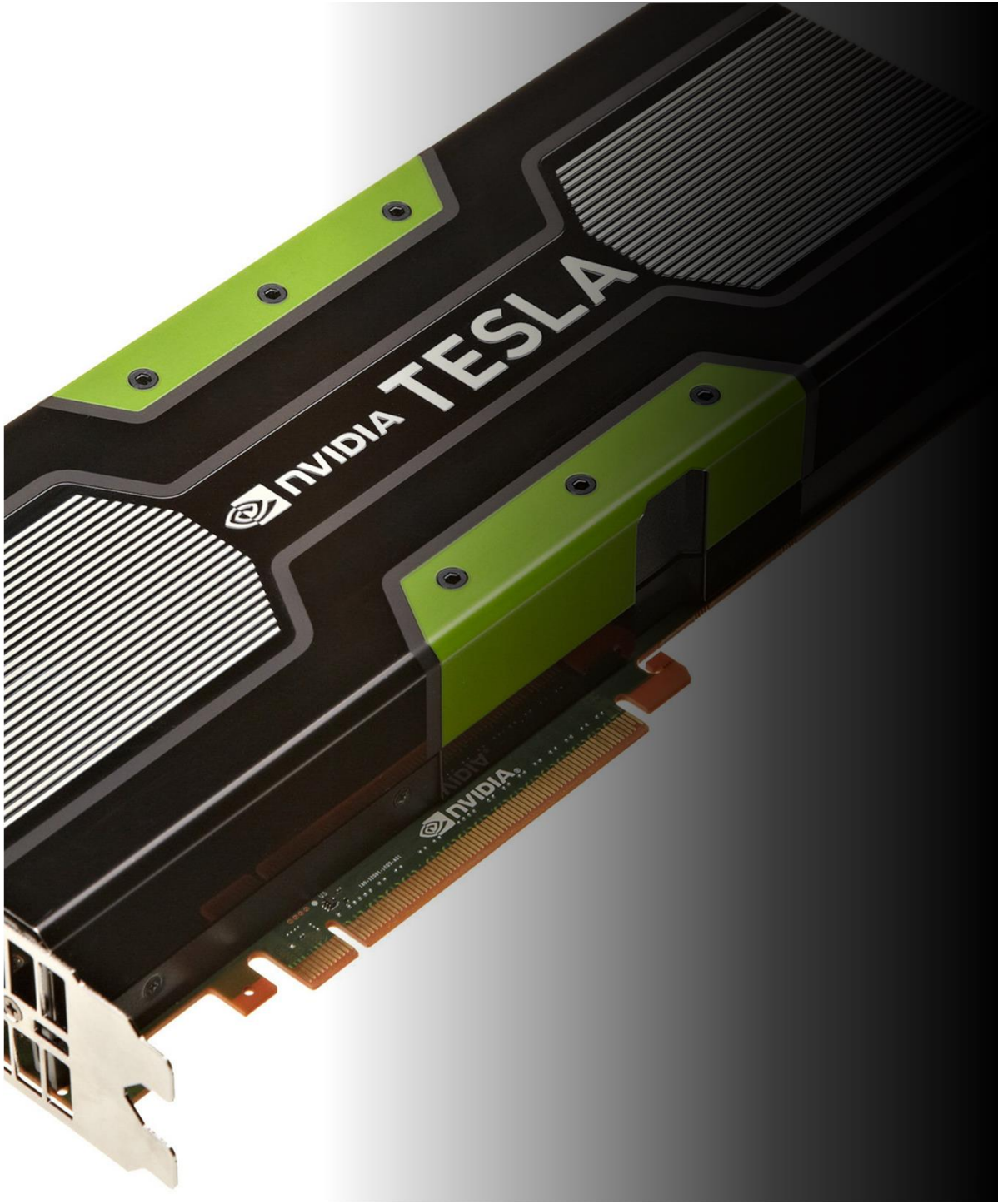
Current Status: We are currently in the era of **Narrow AI**. **AGI** is an active area of research, while **ASI** remains a theoretical concept.



The History of AI



Source: https://qbi.uq.edu.au/brain/intelligent-machines/history-artificial-intelligence + additions



What accounts for recent successes in AI?

- Dominance of machine learning.
- Faster computers and specialized hardware (GPUs).
- Lots of data (the Internet, text, sensors) and storage (cloud)
- New optimization methods (deep learning).

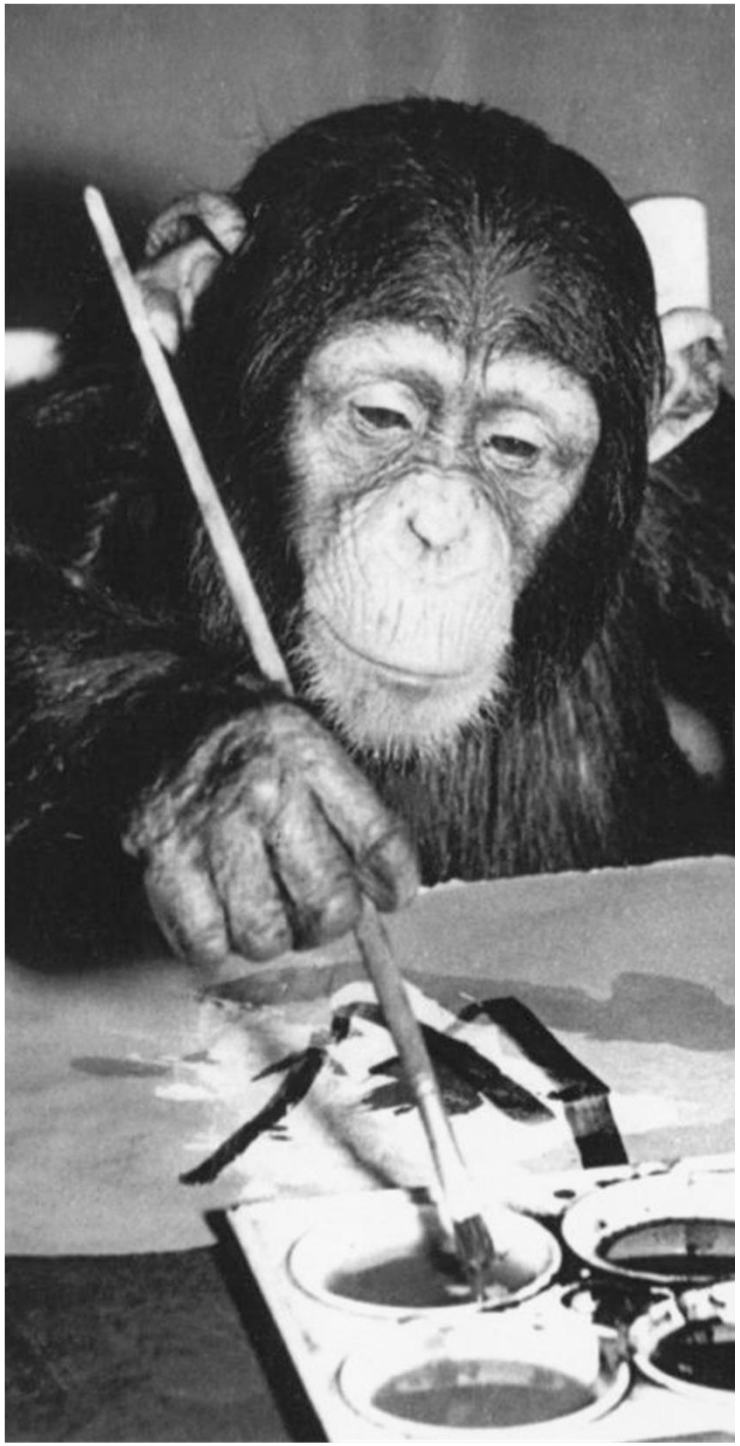
“Moravec’s Paradox”

Hans Moravec (1988): *“It is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and* ***difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility****.”*

A teenager can learn how to drive in a few hours with very little input, but we still have no truly self-driving car.



https://www.newsweek.com/googles-new-two-legged-robot-future-warfare-429831

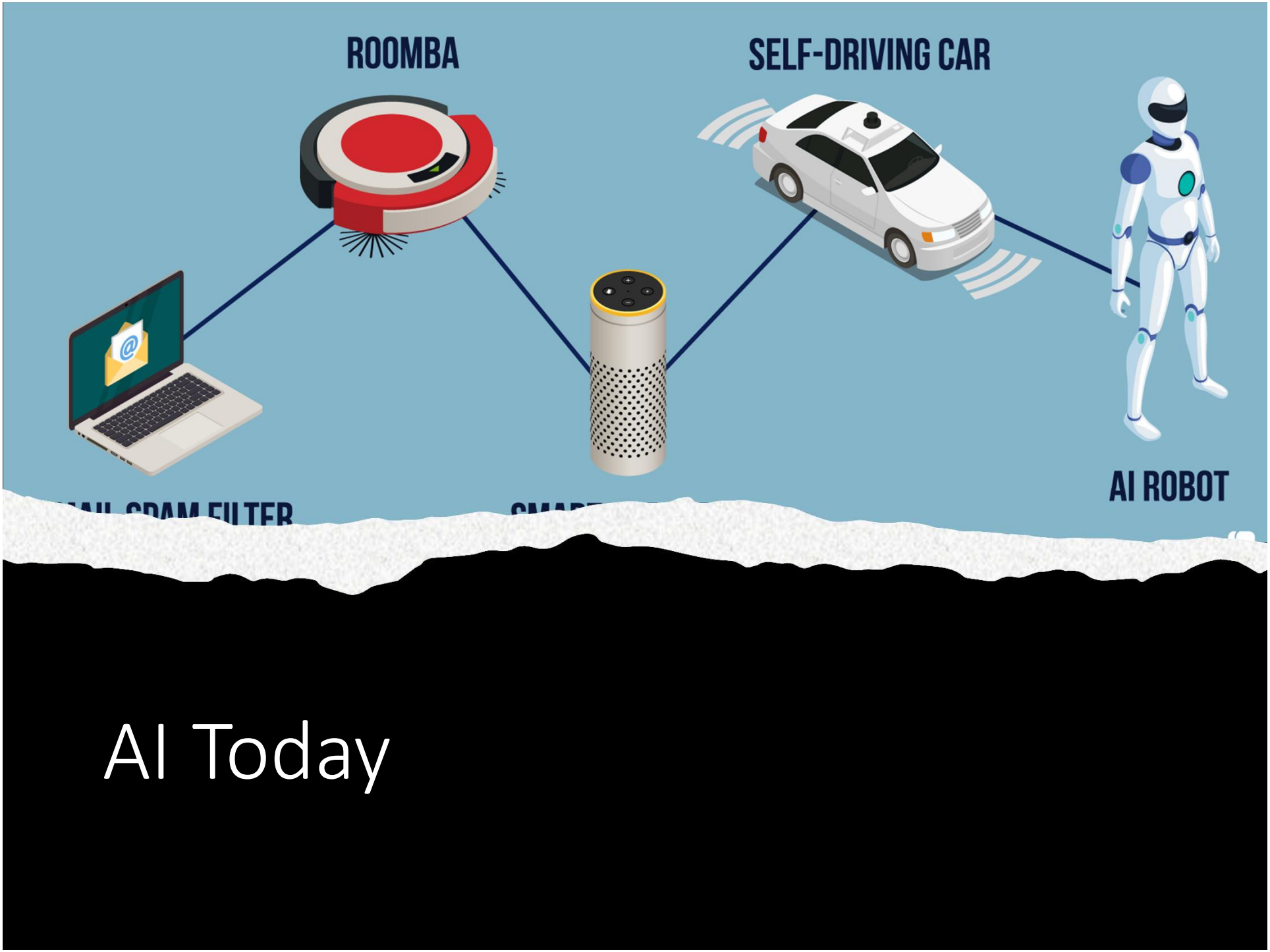


The AI Effect: AI gets no respect?

As soon as a machine gets good at performing some task, the task is no longer considered to require much intelligence.

- **Calculating** ability used to be prized – not anymore.
- **Chess** was thought to require high intelligence – now computers play at a super-human level.
- **Learning** once thought uniquely human - now machine learning is a well-developed discipline.
- **Art**? “Even a monkey can do this!”

What about Large Language Models and Generative AI?

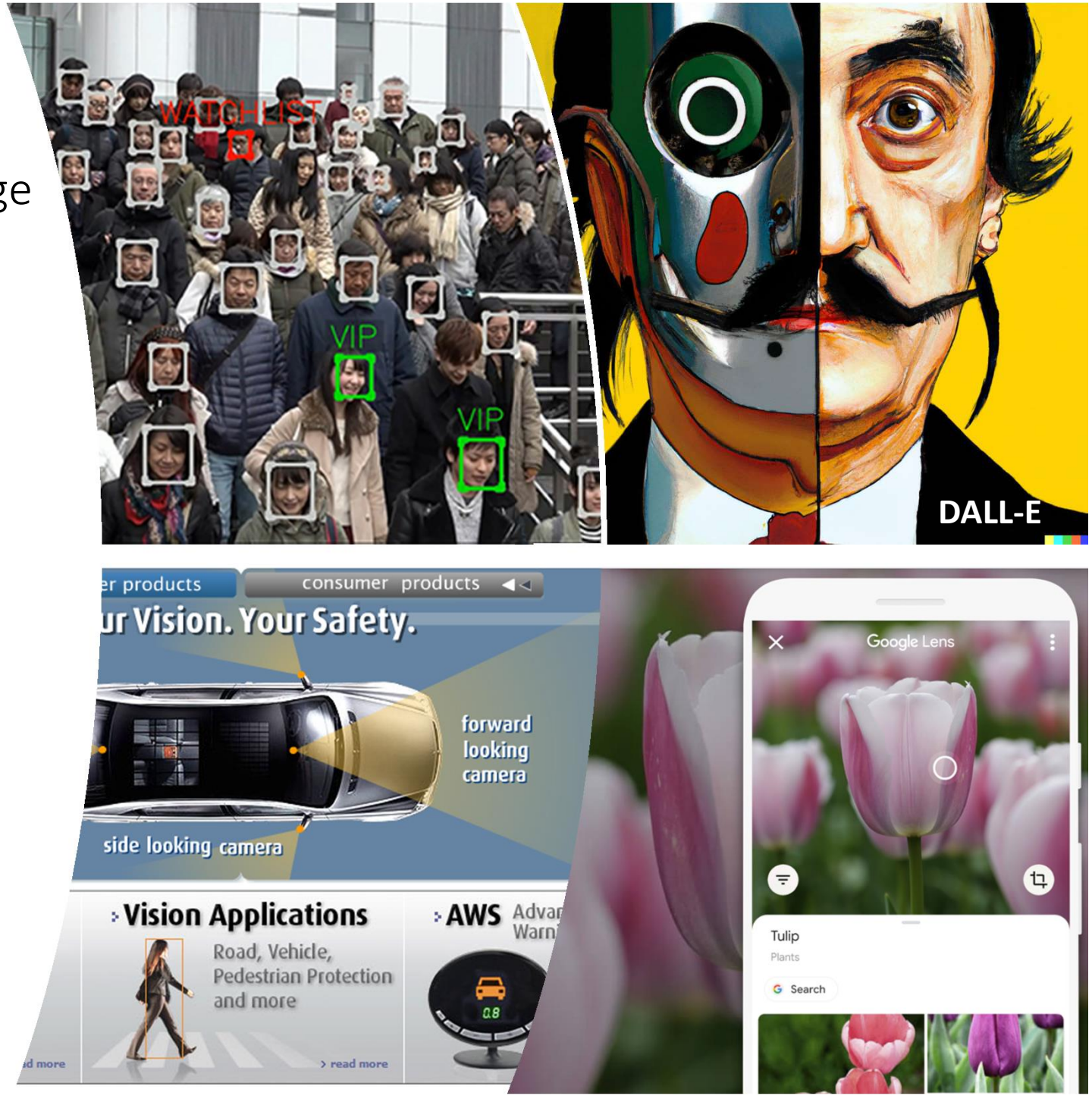


AI Today

Vision and Image Processing

- **OCR**: read license plates, handwriting recognition (e.g., mail sorting).
- **Face detection**: now standard for smart phone cameras.
- **Vehicle safety systems**
- **Visual search**
- **Image generation**

All these technologies operate now at superhuman performance.

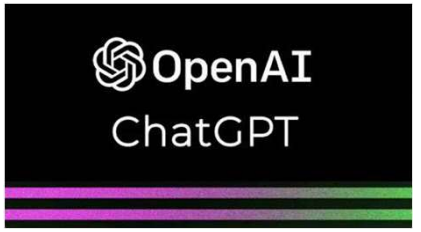


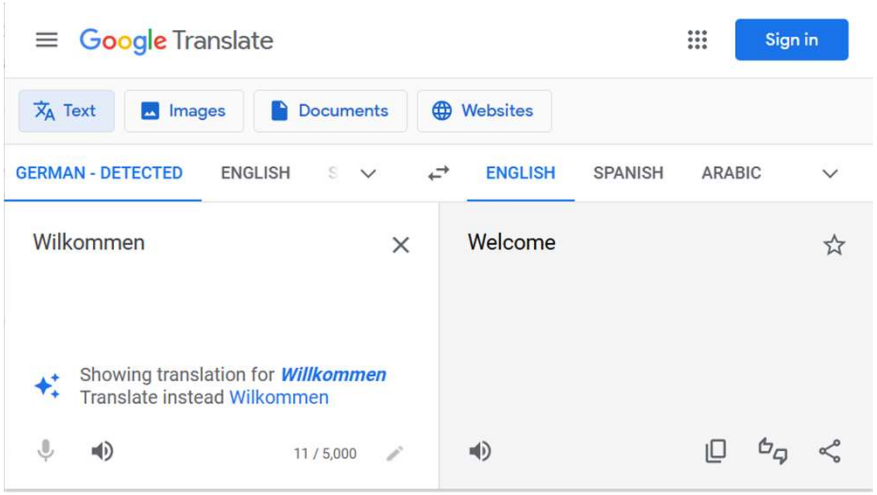
Natural Language Processing



- Text-to-speech
- Speech-to-text to detect voice commands
- Machine translation
- Text generation (Q/A systems) using Large Language Models







These technologies operate now with close to or even superhuman performance.

Humans use language to reason. Does that mean AI that can create high-quality text can reason?

Language understanding is still elusive!

Robotics

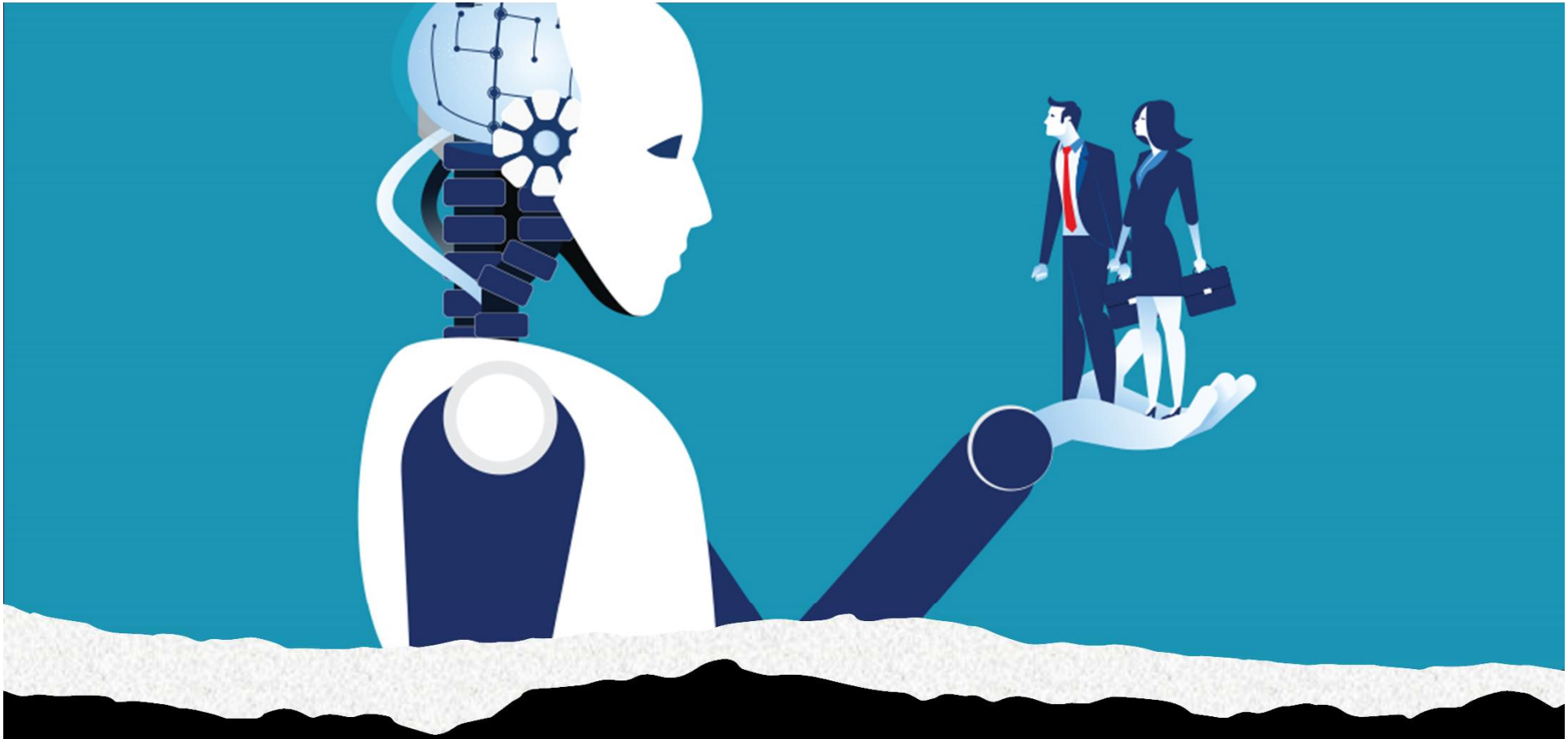
- Mars rovers
- Robot soccer
  - RoboCup
- Autonomous vehicles
  - DARPA Grand Challenge
  - Self-driving cars
- Drones
- Personal robotics
  - Humanoid robots
  - Robotic pets
  - Personal assistants?







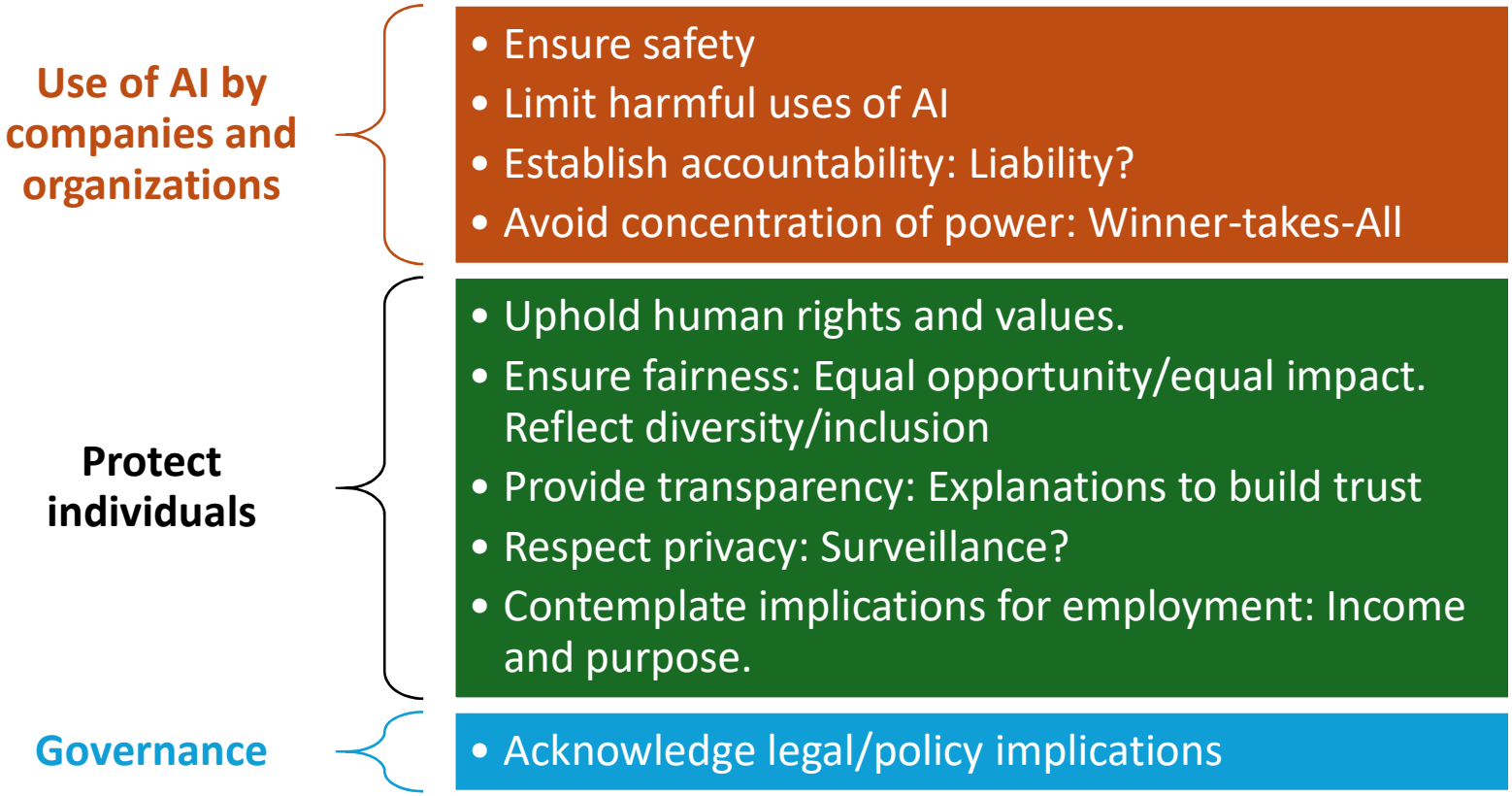




AI Ethics & Safety

A new Frontier for
Fairness and Freedom
AIMA Chapter 27

Commonly-Cited Safety and Ethics Principles



Next, we look at the implementation of these principles in different countries.

European Union

Has regulations since 2016 included in the General Data Protection Regulation (GDPR)

Art. 22 GDPR – Automated individual decision-making, including

California Consumer Privacy Act

California's CCPA was not modeled after the GDPR

Art. 22 GDPR

2016

Automated individual decision-making, including profiling

1. The data subject shall have the right not to be subject to a decision based solely on automated processing, including profiling, which produces legal effects concerning him or her or similarly significantly affects him or her.

2. Paragraph 1 shall not apply if the decision:

   (a) is necessary for entering into, or performance of, a contract between the data subject and a data controller;

   (b) is authorised by Union or Member State law to which the controller is subject and which also lays down suitable measures to safeguard the data subject's rights and freedoms and legitimate interests; or

   (c) is based on the data subject's explicit consent.

3. In the cases referred to in points (a) and (c) of paragraph 2, the data controller shall implement suitable measures to safeguard the data subject's rights and freedoms and legitimate interests, at least the right to obtain human intervention on the part of the controller, to express his or her point of view and to contest the decision.

4. Decisions referred to in paragraph 2 shall not be based on special categories of personal data referred to in Article 9(1), unless point (a) or (g) of Article 9(2) applies and suitable measures to safeguard the data subject's rights and freedoms and legitimate interests are in place.

In the United States

Bill Introduced was 2019
Did not receive a vote in Congress.

116TH CONGRESS
1ST SESSION

H. R. 2231

To direct the Federal Trade Commission to require entities that use, store, or share personal information to conduct automated decision system impact assessments and data protection impact assessments.

(2) AUTOMATED DECISION SYSTEM IMPACT ASSESSMENT.—The term "automated decision system impact assessment" means a study evaluating an automated decision system and the automated decision system's development process, including the design and training data of the automated decision system, for impacts on accuracy, fairness, bias, discrimination, privacy, and security that includes, at a minimum—

(A) a detailed description of the automated decision system, its design, its training, data, and its purpose;

(B) an assessment of the relative benefits and costs of the automated decision system in light of its purpose, taking into account relevant factors, including—

(i) data minimization practices;

(ii) the duration for which personal information and the results of the automated decision system are stored;

(iii) what information about the automated decision system is available to consumers;

(iv) the extent to which consumers have access to the results of the automated decision system and may correct or object to its results; and

(v) the recipients of the results of the automated decision system;

(C) an assessment of the risks posed by the automated decision system to the privacy or security of personal information of consumers and the risks that the automated decision system may result in or contribute to inaccurate, unfair, biased, or discriminatory decisions impacting consumers; and

(D) the measures the covered entity will employ to minimize the risks described in subparagraph (C), including technological and physical safeguards.

European Union Study

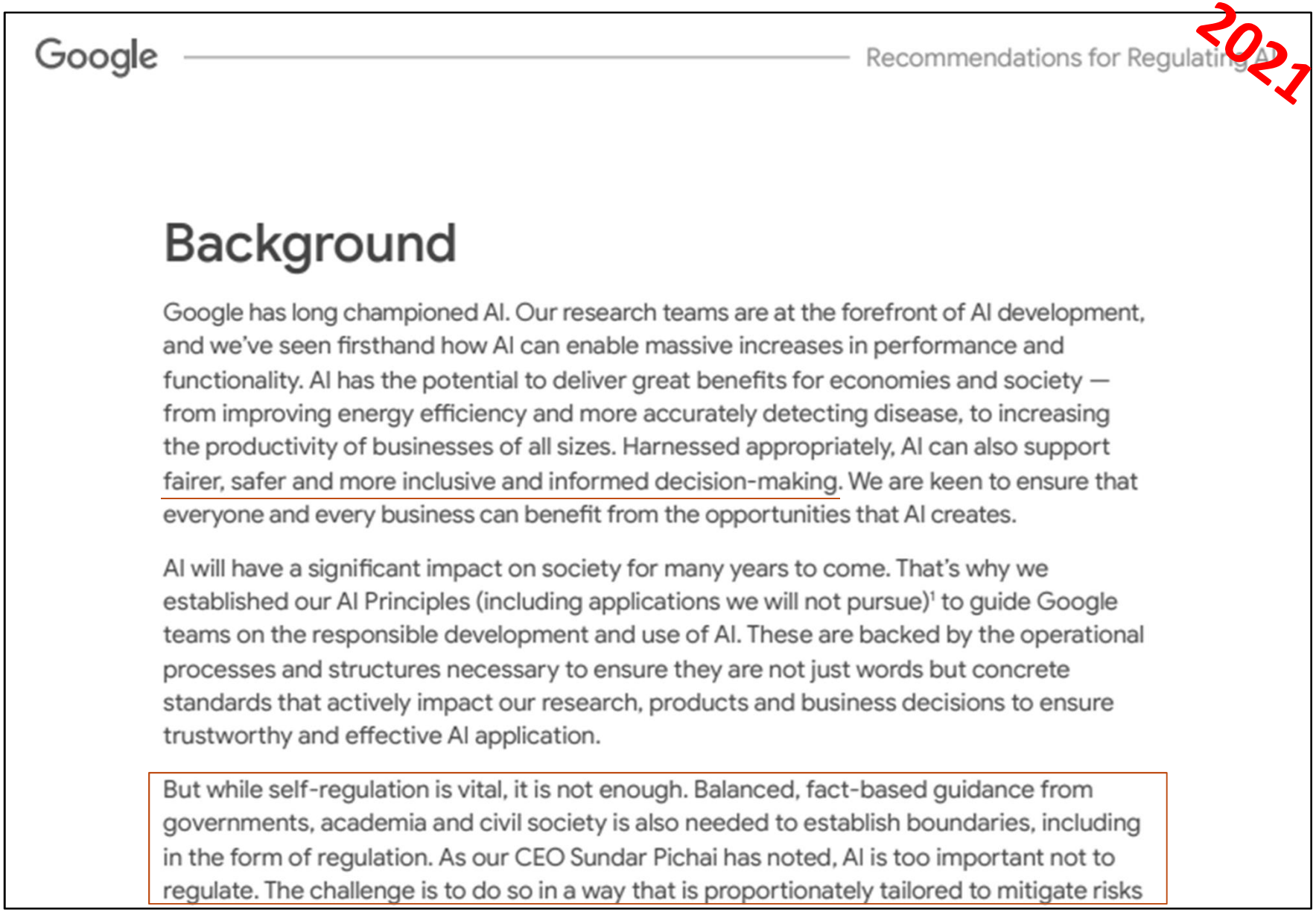


2019

A governance framework for algorithmic accountability and transparency

This study develops policy options for the governance of algorithmic transparency and accountability, based on an analysis of the social, technical and regulatory challenges posed by algorithmic systems. Based on a review and analysis of existing proposals for governance of algorithmic systems, a set of four policy options are proposed, each of which addresses a different aspect of algorithmic transparency and accountability: 1. awareness raising: education, watchdogs and whistleblowers; 2. accountability in public-sector use of algorithmic decision-making; 3. regulatory oversight and legal liability; and 4. global coordination for algorithmic governance.

https://www.europarl.europa.eu/thinktank/en/document.html?reference=EPRS_STU(2019)624262



Source: https://ai.google/static/documents/recommendations-for-regulating-ai.pdf, 2021



Administration Priorities The Record

US White House Executive Order 14110

OCTOBER 30, 2023

Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence

BRIEFING ROOM › PRESIDENTIAL ACTIONS

By the authority vested in me as President by the Constitution and the laws of the United States of America, it is hereby ordered as follows:

Section 1. Purpose. Artificial intelligence (AI) holds extraordinary potential for both promise and peril. Responsible AI use has the potential to help solve urgent challenges while making our world more prosperous, productive, innovative, and secure. At the same time, irresponsible use could exacerbate societal harms such as fraud, discrimination, bias, and disinformation; displace and disempower workers; stifle competition; and pose risks to national security. Harnessing AI for good and realizing its myriad benefits requires mitigating its substantial risks. This endeavor demands a society-wide effort that includes government, the private sector, academia, and civil society.

2023

Some important points:

- Artificial Intelligence must be **safe and secure**.
- Promoting **responsible innovation, competition, and collaboration**
- Americans' **privacy, civil liberties and labor rights** must be protected.

January 2025: This Executive Order is now revoked.

Fairness: Algorithmic Bias

"**Algorithmic bias** describes systematic and repeatable errors in a computer system that create unfair outcomes, such as privileging one arbitrary group of users over others." [Wikipedia]



Pre-existing bias

Social and institutional norms influence design and training data choices.

Example: Evaluate job applicants for a job which is historically almost exclusively held by males.



Technical bias

Limitations of a program or computational power.

Example: Algorithm uses data that is easy to obtain but biased.



Emergent bias

Use and reliance on algorithms across new or unanticipated contexts.

Example: Use a large language model to support job candidate selection.

Types of AI Safety

"Prevent accidents, misuse, or other harmful consequences of AI."



AI Testing



Monitoring AI



Adversarial robustness

How should this be ensured?

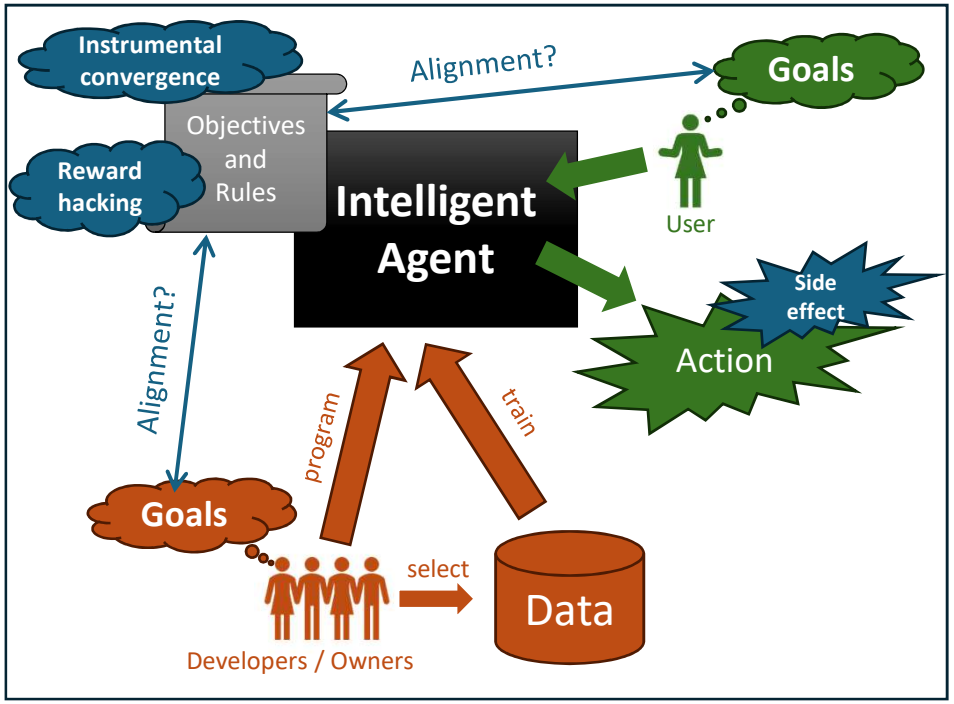
- Corporate self-regulation
- Private watchdogs
- Government action
- International treaties

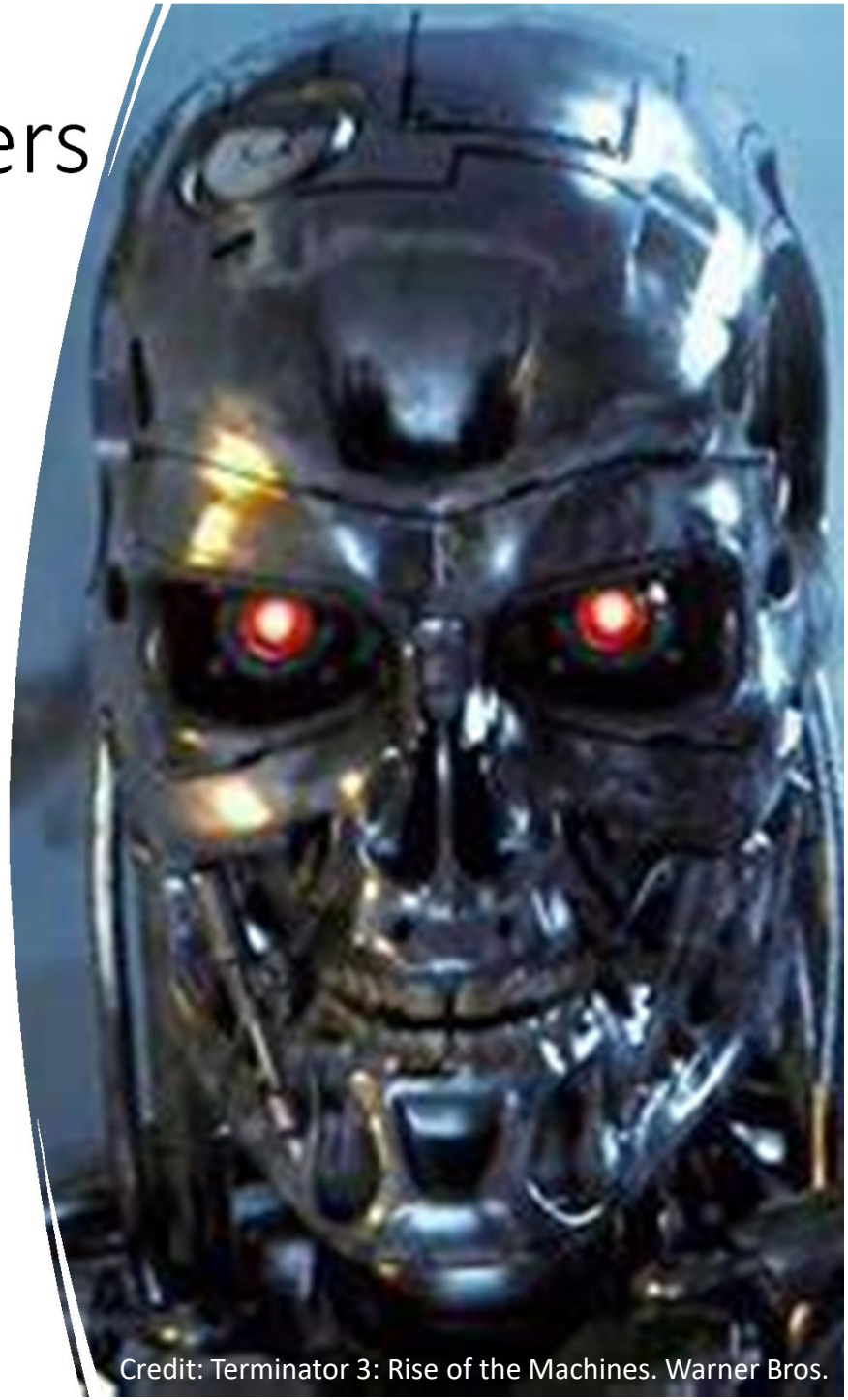
Is it reasonable to assume that a superintelligent AI can get around being tested and monitored?

AI Safety and Optimizers

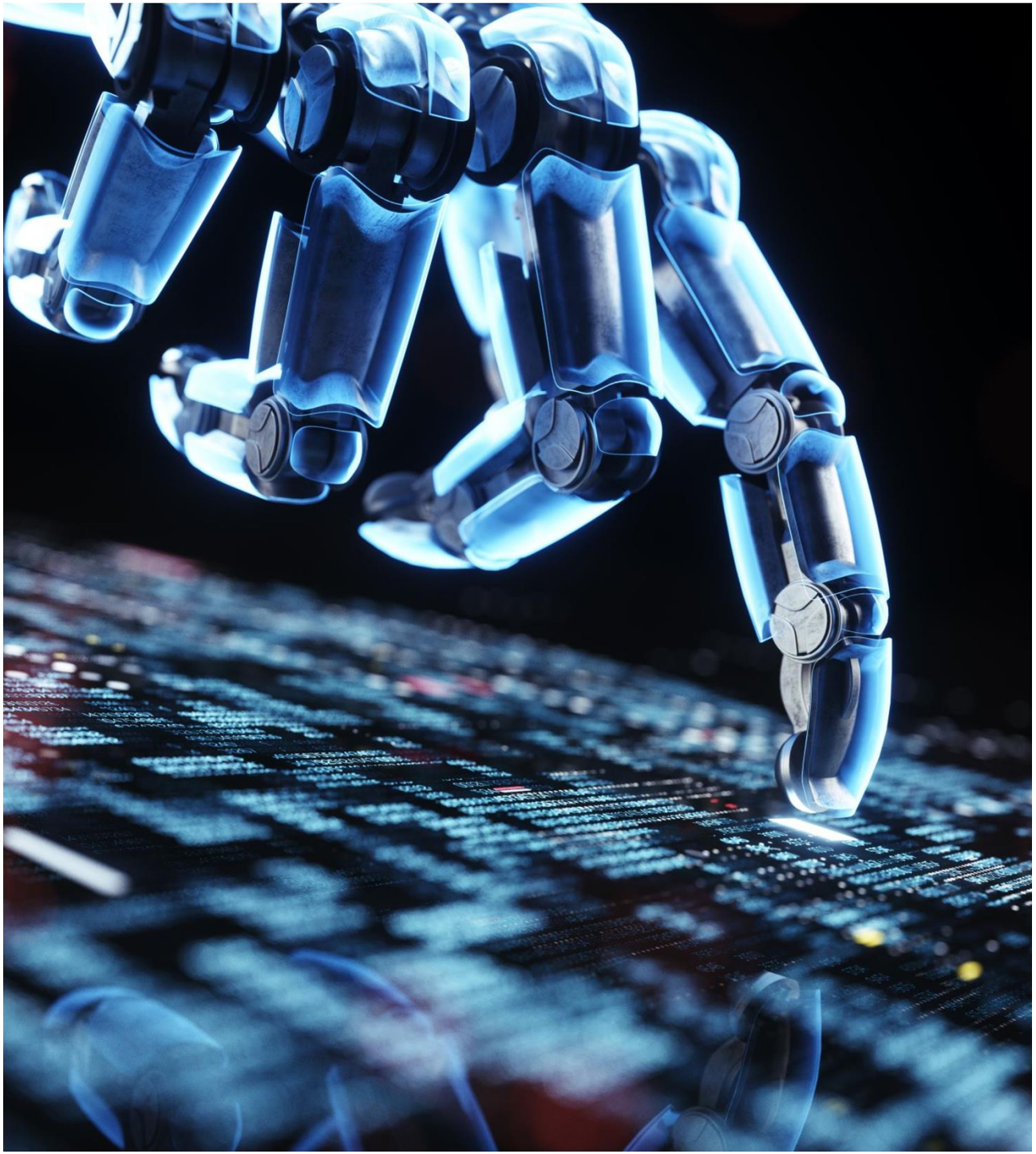
Intelligent Agents are “optimizers!”

- **Goal/reward alignment**: How do we specify a robust objective function? Whose objectives are used?
- **Reward hacking:** The AI learns to exploit unintended side effects to get a high “score” without solving the objective. AI needs to follow social norms.
- **Instrumental convergence**: All intelligent agents will pursue common subgoals like the need for more power to get better at reaching its objectives. How will this need be balanced with human’s needs?





Credit: Terminator 3: Rise of the Machines. Warner Bros.



Outlook

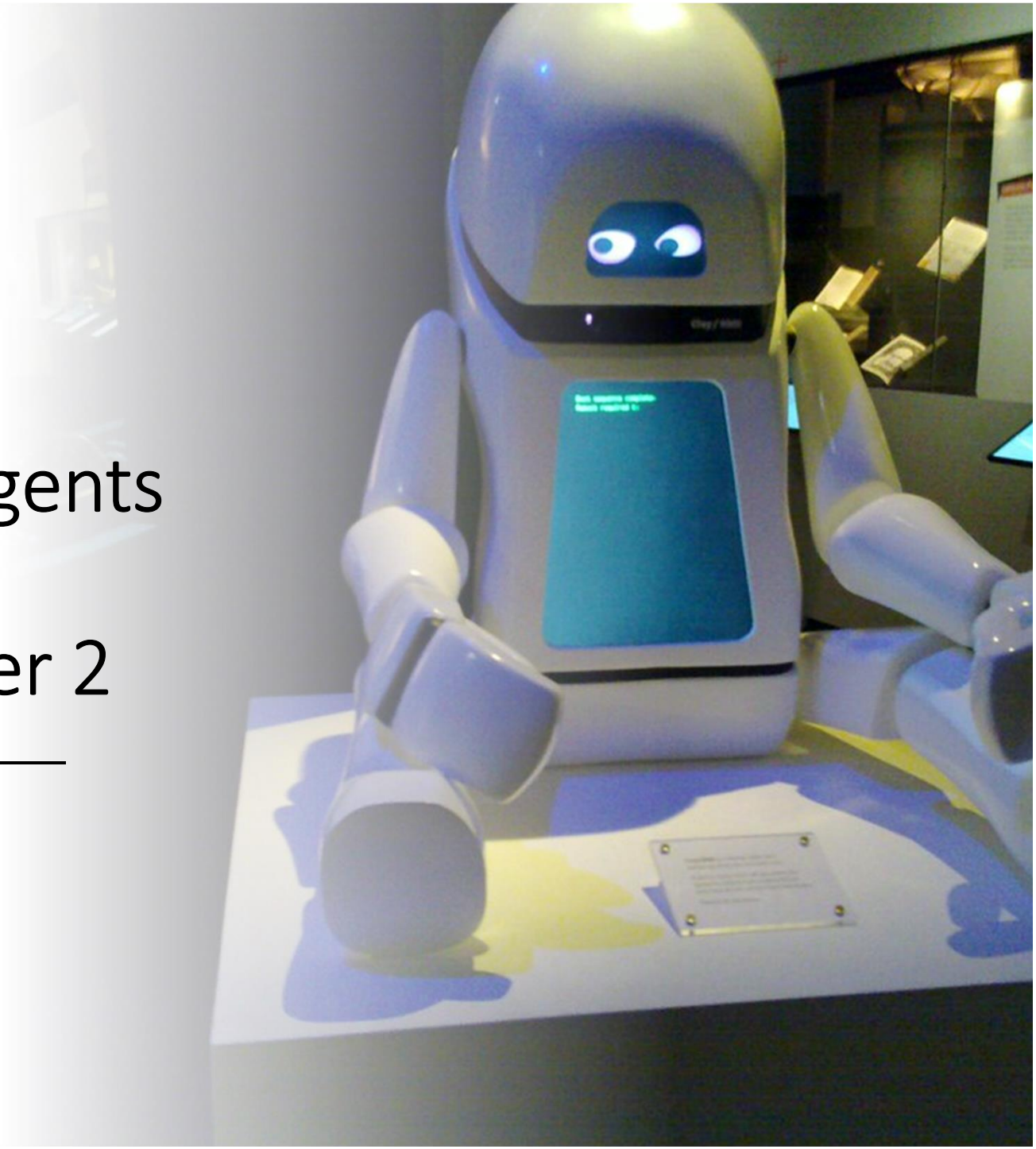
AI is a technology that is on the verge of significant leaps…

- New technologies always had a **profound impacted** on the way we live and work (e.g., electricity, the internet, mobile communication).
- We can expect unprecedented gains in productivity from better **narrow AI**.
- New technologies always also present **dangers** and need to be regulated.

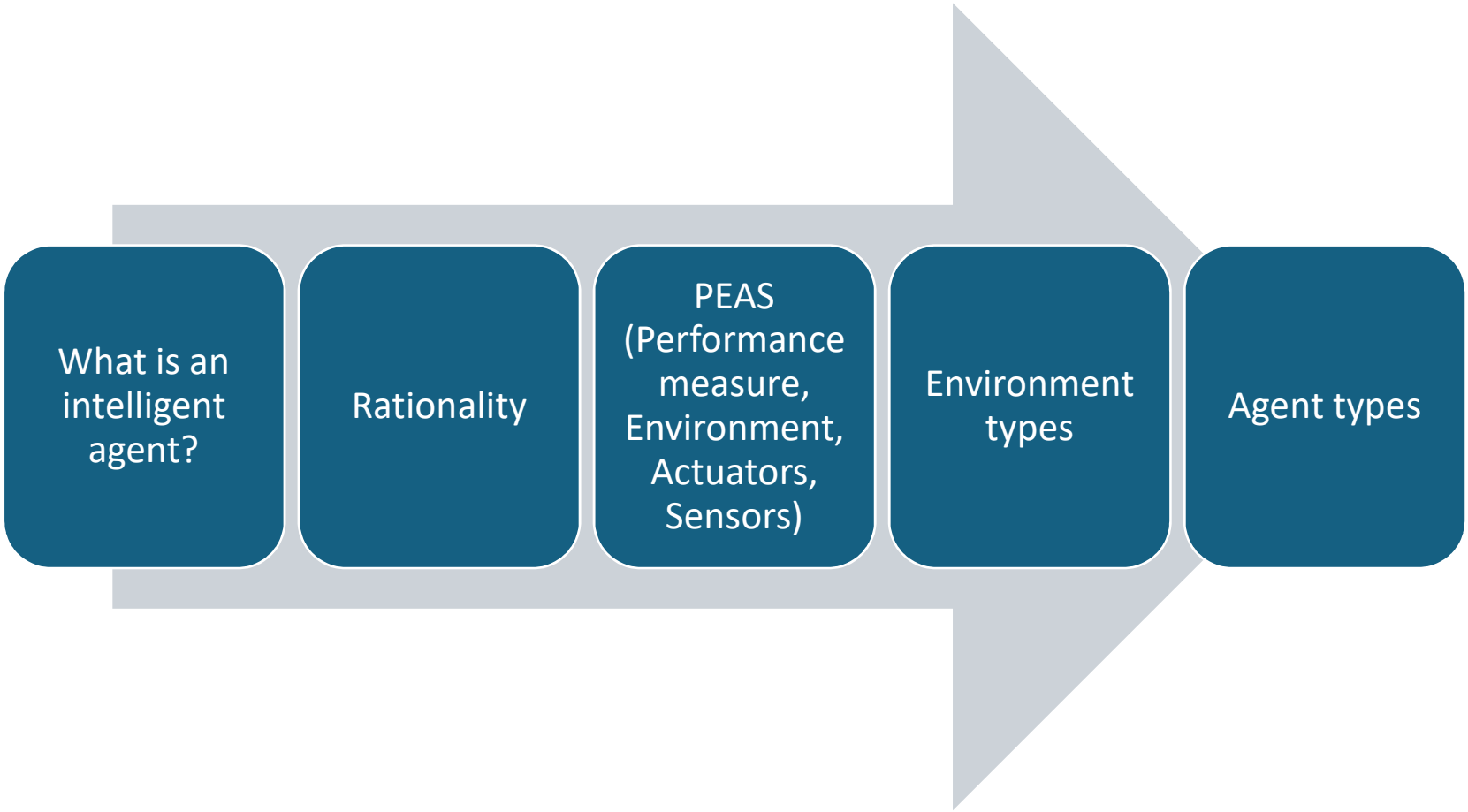
This course will introduce simple techniques to create intelligent agents.

Intelligent Agents

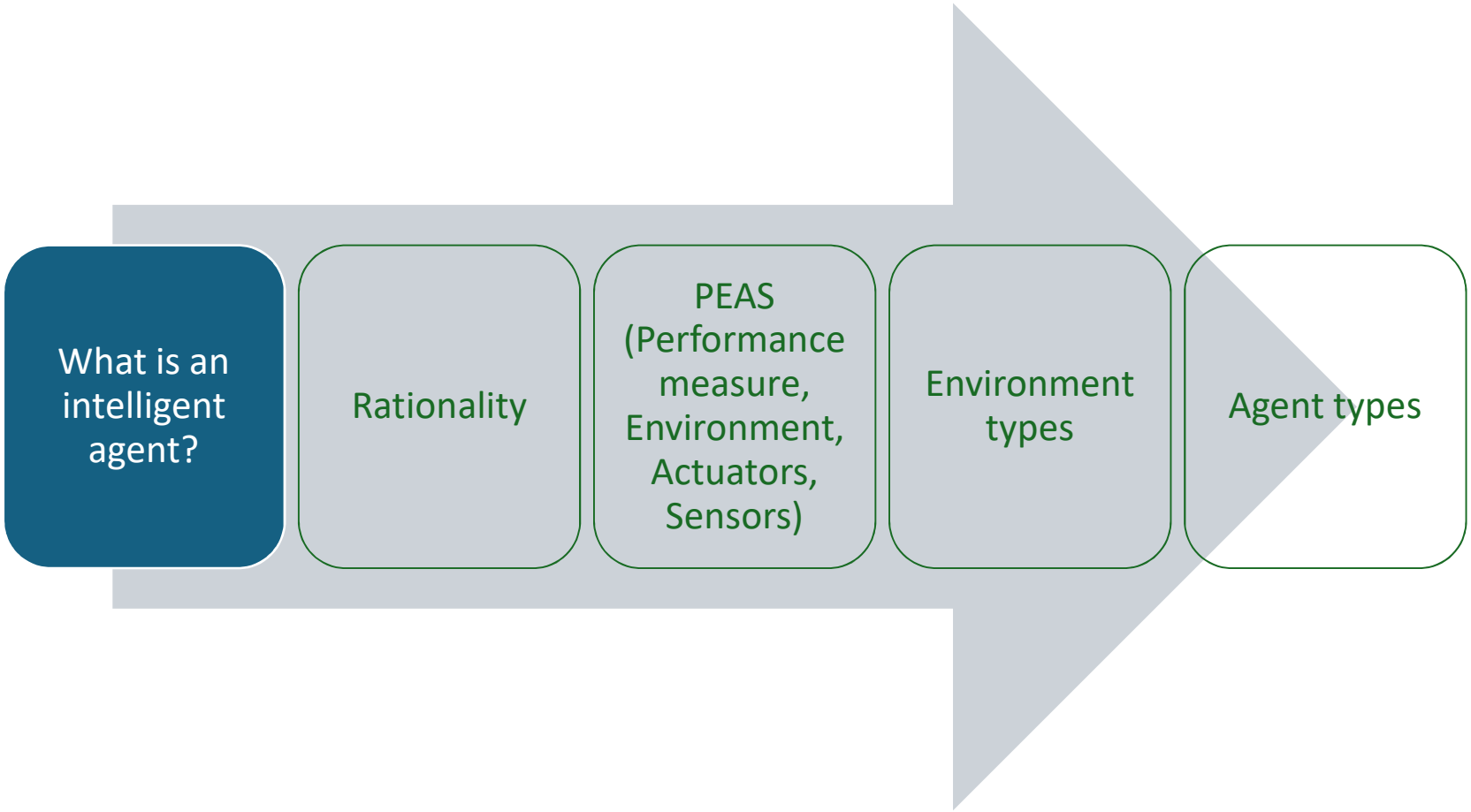
AIMA Chapter 2



Outline

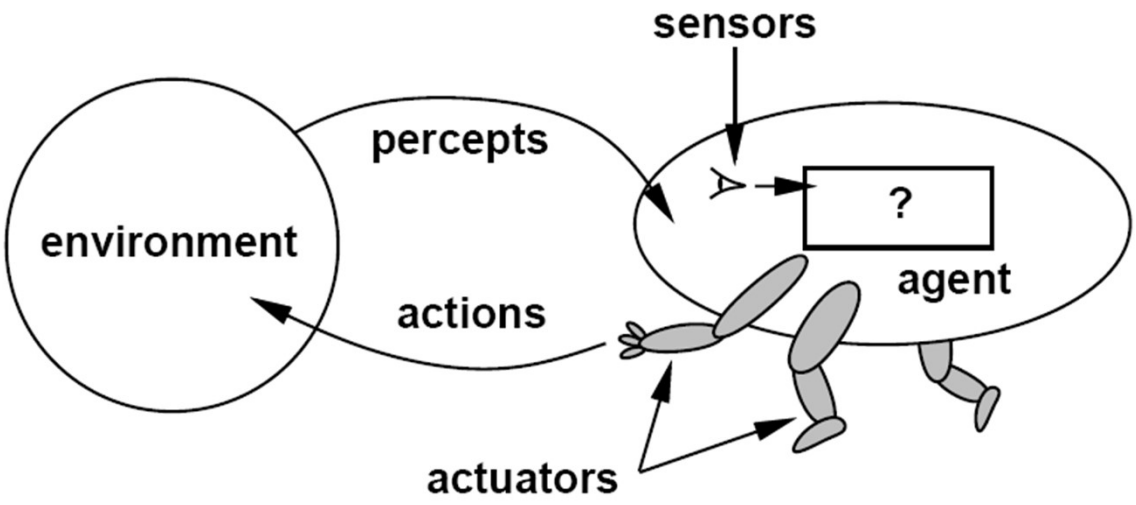

What is an intelligent agent?
Rationality
PEAS (Performance measure, Environment, Actuators, Sensors)
Environment types
Agent types

Outline: What is an Intelligent Agent



What is an Agents?

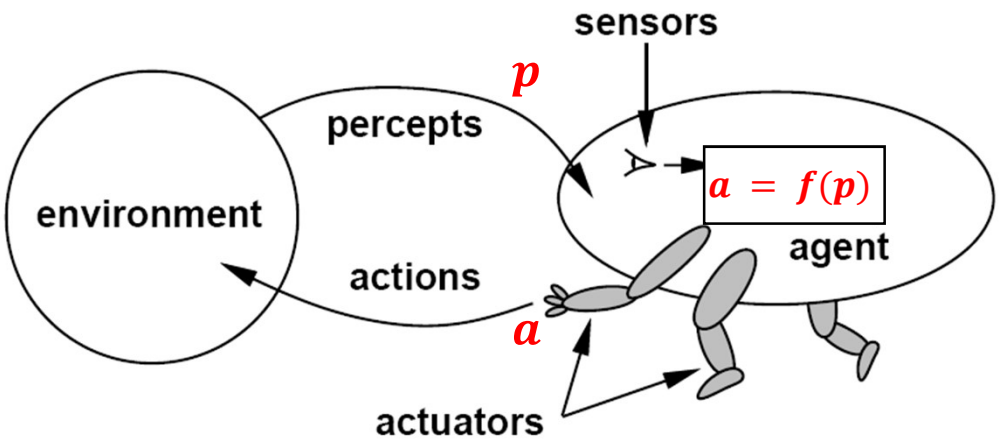
- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.



Agent Function and Agent Program

The agent function maps from the set of all possible *percept sequences* P^* to the *set of actions* A formulated as an abstract mathematical function.

$$f : P^* \rightarrow A$$

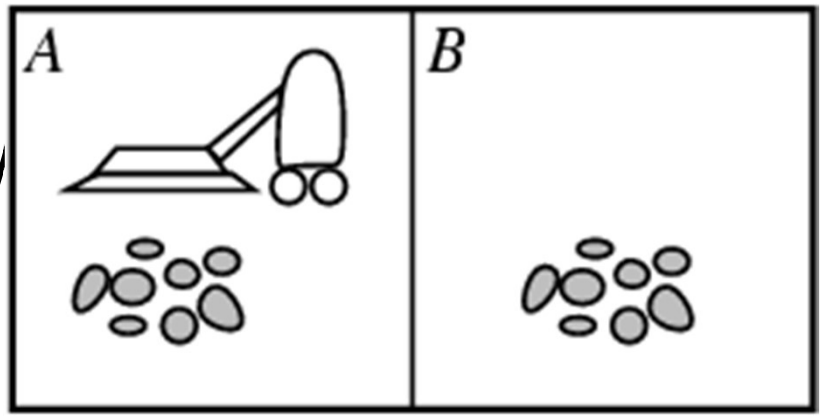


The agent program is a concrete implementation of this function for a given physical system.

Agent = architecture (hardware) + agent program (implementation of f)

- Sensors
- Memory
- Computational power

Example: Vacuum-cleaner W



- **Percepts:**
  Location and status, e.g., [A, Dirty]
- **Actions:**
  Left, Right, Suck, NoOp

Agent function: $f : P^* \rightarrow A$

Percept Sequence	Action
[A, Dirty]	Suck
[A, Clean]	Right
…	
[A, Dirty][A, Clean], [B, Clean]	Left
…	
[A, Dirty][A, Clean], [B, Clean], [A, Dirty]	Suck
…	

Problem: This table can become infinitively large!

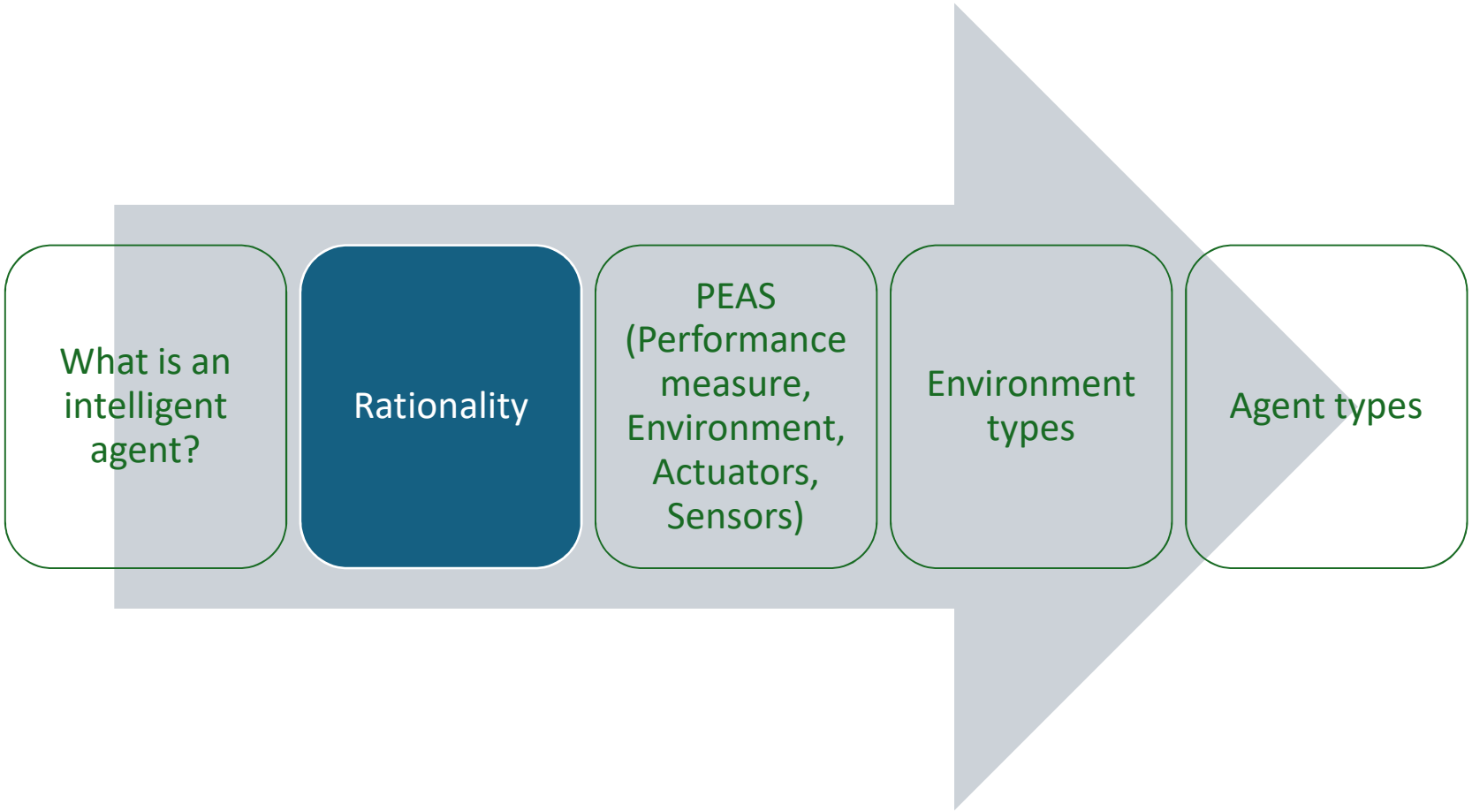
Most recent Percept p

Implemented agent program:

function Vacuum-Agent([location, status])
returns an action a

```
if status = Dirty then return Suck
else if location = A then
                        return Right
else if location = B then
                        return Left
```

Outline: Rationality



Rational Agents: What is Good Behavior?

Foundation from normative moral theory and economics:

- **Consequentialism**: Evaluate actions by their consequences.
- **Utilitarianism**: Maximize happiness and well-being.

Definition of a rational agent:

*"For each possible percept sequence, a rational agent should select an **action** that **maximizes its expected performance measure**, given the evidence provided by the **percept sequence** and the **agent's built-in knowledge**."*

- **Performance measure**: An *objective* criterion for success of an agent's behavior (often called utility function or reward function).
- **Expectation**: Outcome averaged over all possible situations that may arise.

Rule: Pick the action that maximize the expected utility

$$a = \text{argmax}_{a \in \text{A}} \, E(U \mid a)$$

Rational Agents

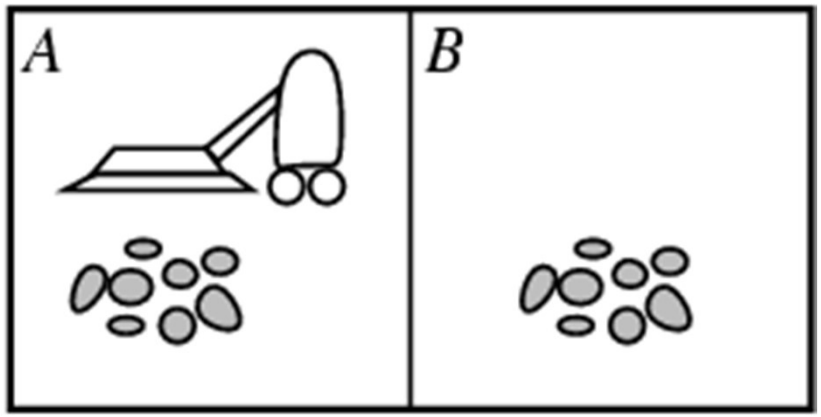
Rule: Pick the action that maximize the expected utility

$$a = \text{argmax}_{a \in \text{A}} \, E(U \mid a)$$

This means:

- **Rationality is an ideal** – it implies that no one can build a better agent
- **Rationality ≠ Omniscience** – rational agents can make mistakes if percepts and knowledge do not suffice to make a good decision
- **Rationality ≠ Perfection** – rational agents maximize **expected** outcomes not actual outcomes
- **It is rational to explore and learn** – I.e., **use percepts** to supplement prior knowledge and become autonomous
- **Rationality is often bounded** by available memory, computational power, available sensors, etc.

Example: Performance Measure for the Vacuum-cleaner World



- **Percepts:**
  Location and status, e.g., [A, Dirty]
- **Actions:**
  Left, Right, Suck, NoOp

Agent function:

Percept Sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
…	
[A, Clean], [B, Clean]	Left
…	

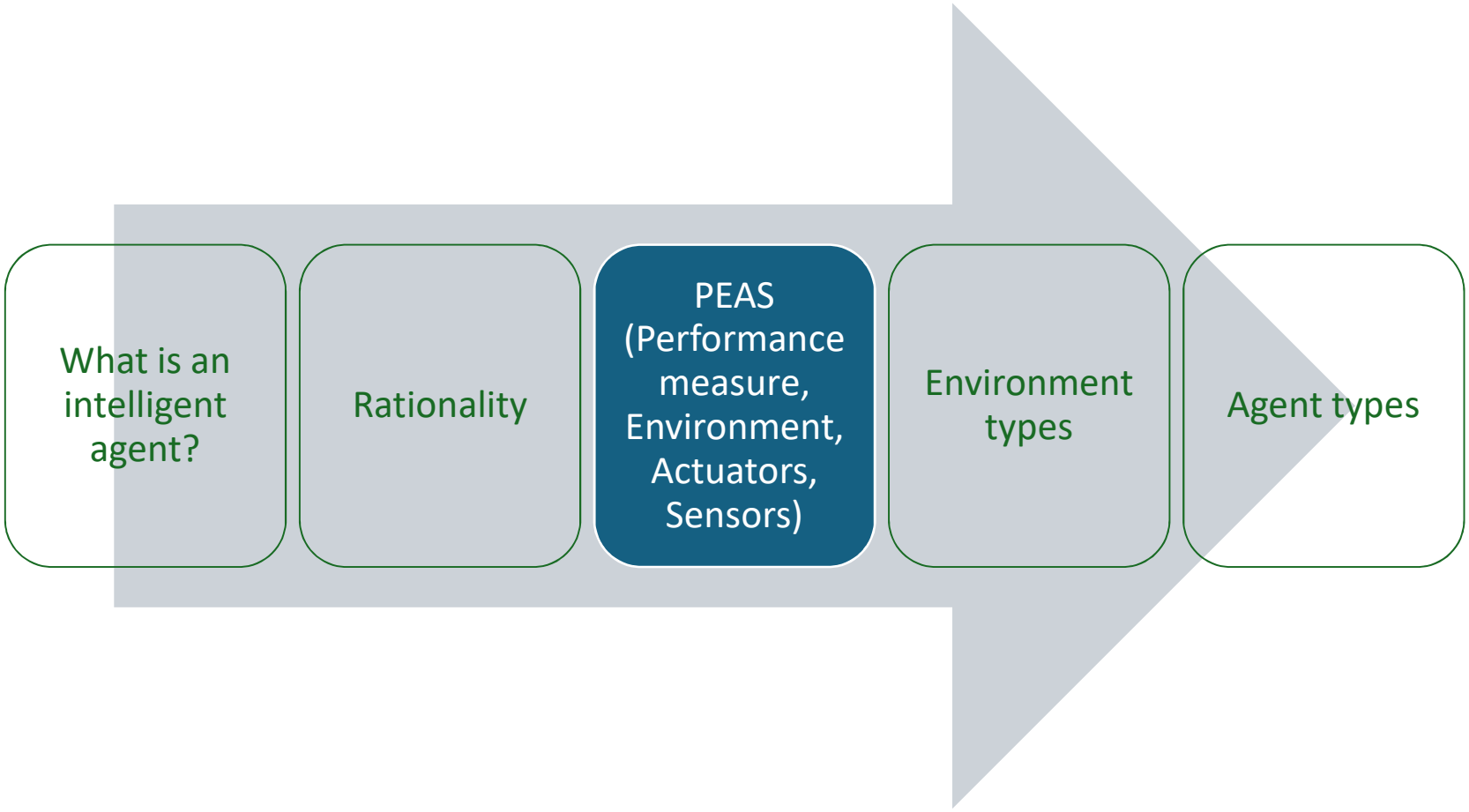
Implemented agent program:

function Vacuum-Agent([location, status])
returns an action

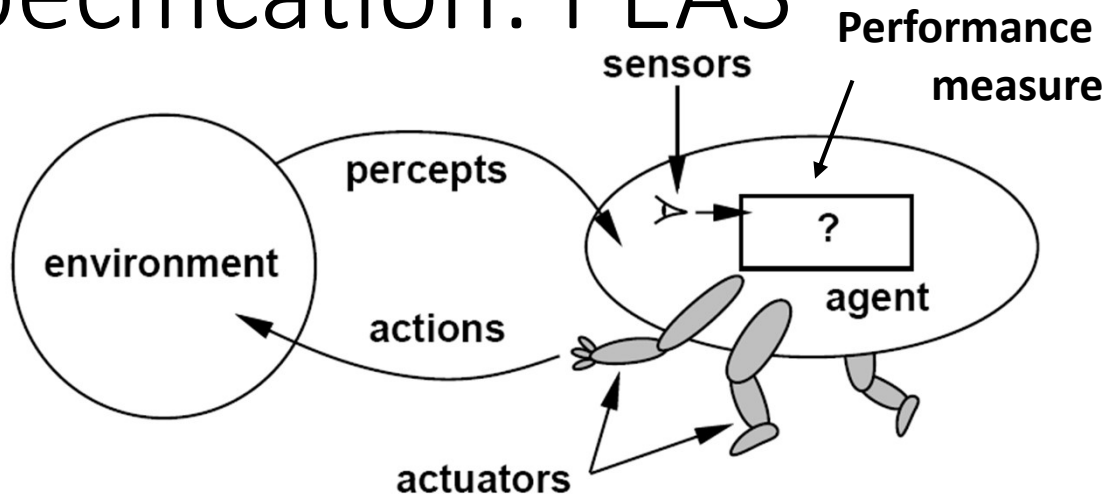
```
if status = Dirty then return Suck
else if location = A then return Right
else if location = B then return Left
```

What could be a performance measure?
Is this agent program rational?

Outline: PEAS



Problem Specification: PEAS



Performance measure	Environment	Actuators	Sensors
How do we measure whether the agent is successful?	Where does the agent operate?	How does the agent affect the environment ?	How does the agent perceive the environment ?

Why do we need PEAS?

- Suppose someone says:

  "**Build an AI robot.**"

- This statement is incomplete.
- You should immediately ask four questions:
  - What should it achieve? (**Performance Measure**)
  - Where will it work? (**Environment**)
  - How will it perform actions? (**Actuators**)
  - How will it obtain information? (**Sensors**)
- Without answering these questions, an intelligent agent cannot be properly designed.

Performance Measure (P)

A **Performance Measure** specifies **how success is evaluated**.

- **Example 1: Vacuum Cleaner**
- Should the robot
  - move fast?
  - clean everything?
  - consume less electricity?
- The designer decides.
- Possible performance measures:
  - Maximum clean squares
  - Minimum time
  - Minimum electricity
  - Minimum collisions

Environment (E)

The environment is

- Everything outside the agent that interacts with it.
- Think of the environment as the world in which the agent lives.

- Examples
  - Vacuum robot
- Environment
  - Rooms
  - Walls
  - Dirt
  - Furniture

Actuators (A)

- Actuators are
  - The mechanisms through which an agent performs actions.
- Think
  - Sensors = Input
  - Actuators = Output
- For Vacuum Cleaner Agent, Wheel motors, vacuum motor, brushes

Sensors (S)

- Sensors are
  - Devices used to perceive the environment.
- Without sensors,
  - the agent becomes blind.
- For Vacuum Cleaner Agent, Dirt sensor, location sensor, bump sensor, cliff sensor, camera/LiDAR, battery sensor

Example: Automated Taxi Driver

Performance measure	Environment	Actuators	Sensors
• Safe • fast • legal • comfortable trip • maximize profits	• Roads • other traffic • pedestrians • customers	• Steering wheel • accelerator • brake • signal • horn	• Cameras • sonar • speedometer • GPS • Odometer • engine sensors • keyboard

Example: Spam Filter

Performance measure	Environment	Actuators	Sensors
• Accuracy: Minimizing false positives, false negatives	• A user's email account • email server	• Mark as spam • delete • etc.	• Incoming messages • other information about user's account

Exercise

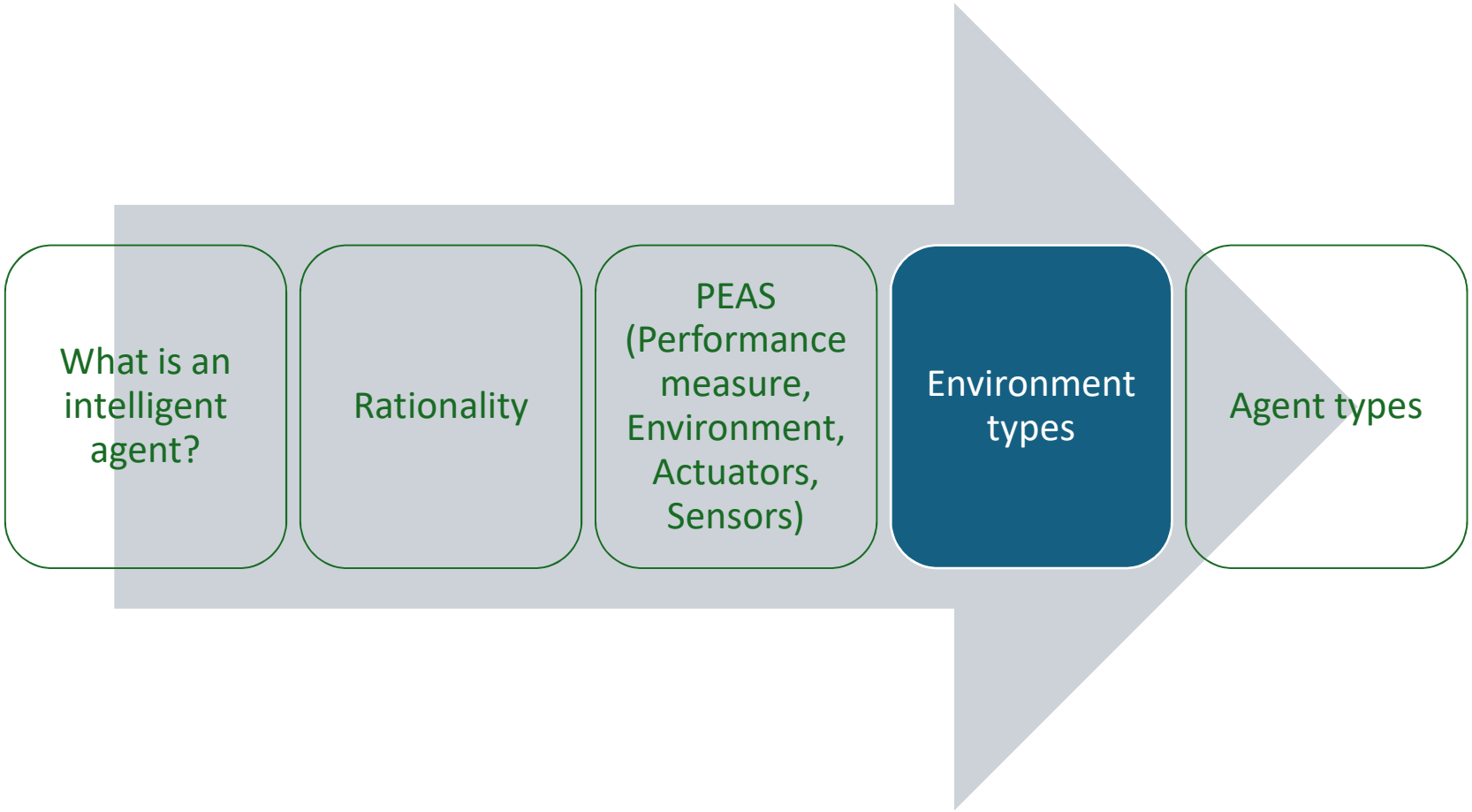
- For each of the following activities, give a PEAS description of the task environment

- Playing soccer.
- Exploring the subsurface oceans of Titan.
- Shopping for used AI books on the Internet.
- Playing a tennis match.
- Practicing tennis against a wall.
- Performing a high jump.
- Knitting a sweater.
- Bidding on an item at an auction.

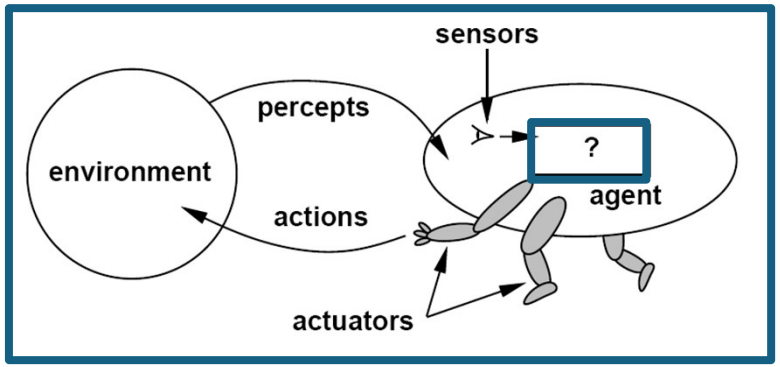
Activity	Performance Measure (P)	Environment (E)	Actuators (A)	Sensors (S)
1. Playing Soccer	Win match, score goals, teamwork, avoid fouls, conserve energy	Soccer field, teammates, opponents, referee, ball, goalposts, weather	Legs, feet, arms (throw-in), head, body movement, voice	Eyes, ears, touch, balance (vestibular), proprioception
2. Exploring the Subsurface Oceans of Titan	Collect scientific data, avoid damage, maximize coverage, conserve power, return data	Titan's ocean, ice layers, rocks, currents, unknown obstacles, extreme temperatures	Thrusters, robotic arms, sampling devices, cameras, communication antenna	Sonar, cameras, pressure sensors, temperature sensors, chemical sensors, IMU, depth sensors
3. Shopping for Used AI Books on the Internet	Buy correct book, minimum price, good condition, reliable seller, fast delivery	Online bookstores, auction sites, sellers, payment systems, Internet	Keyboard, mouse, browser/API requests, payment submission	Web pages, search results, prices, seller ratings, book descriptions
4. Playing a Tennis Match	Win points, win sets, minimize errors, obey rules	Tennis court, opponent, net, ball, umpire, weather	Legs, arms, racket, body movement	Eyes, ears, touch, balance

5. Practicing Tennis Against a Wall	Improve accuracy, consistency, endurance	Wall, tennis court, ball	Legs, arms, racket	Eyes, touch, balance
6. Performing a High Jump	Clear maximum height without knocking bar	Runway, bar, landing mat, weather	Legs, arms, torso	Eyes, balance, proprioception
7. Knitting a Sweater	Correct pattern, minimal mistakes, complete sweater, efficient use of yarn	Yarn, knitting needles, pattern, workspace	Hands, fingers, arms	Eyes, touch
8. Bidding on an Item at an Auction	Win desired item at lowest possible price	Auction platform, competing bidders, seller, auction rules	Mouse, keyboard, bidding software	Current bid, remaining time, bid history, notifications

Outline: Environment Types



The Environment



- We typically consider everything outside the agent function (the agent's brain) as the agent's environment.
- This means that the sensors and actuators are part of the environment.
- The agent function receives already preprocessed percepts and acts by issuing high-level instructions to the actuators.

Examples:







Task Environment Dimensions

- **Fully Observable vs. Partially Observable**: Fully observable means the agent's sensors detect the complete state of the environment at each point in time. Partially observable environments have missing or noisy data due to inaccurate or limited sensors.
- **Single-Agent vs. Multi-Agent**: Single-agent involves an agent operating by itself (e.g., a crossword puzzle). Multi-agent environments include other agents that may act competitively or cooperatively (e.g., chess or driving).

- **Deterministic vs. Stochastic**: Deterministic means the next state of the environment is completely determined by the current state and the action executed by the agent. If there is any uncertainty or randomness, it is stochastic.
- **Episodic vs. Sequential**: Episodic means the agent's experience is divided into atomic chapters; the choice of action in the current episode does not affect future episodes. Sequential environments require looking ahead, as current actions affect future decisions.

- **Static vs. Dynamic**: Static environments do not change while the agent is deliberating. Dynamic environments continuously change, forcing the agent to act quickly. If the environment does not change but the agent's performance score decreases over time, it is called **semidynamic**.
- **Discrete vs. Continuous**: Discrete environments have a finite, distinct number of states, actions, and percepts (e.g., chess). Continuous environments have infinite, smoothly varying possibilities (e.g., taxi driving).
- **Known vs. Unknown**: This refers to the agent's (or designer's) knowledge state about the "laws of physics" of the environment. In a known environment, the outcomes for all actions are given. If the environment is unknown, the agent must learn how it works to make good decisions.

Chess (With a Clock)

- **Fully Observable**: The agent can see the exact position of all pieces on the board at all times.
- **Multi-Agent**: The agent competes against an opponent whose actions directly conflict with its goals.
- **Deterministic**: There is no randomness; a specific move always results in the exact same board configuration.
- **Sequential**: Current moves heavily influence the game state, piece advantages, and strategy 20 moves later.
- **Semidynamic**: The board pieces do not move while the agent thinks, but the clock ticks down, reducing the player's remaining time.
- **Discrete**: There is a finite, countable number of valid moves and board positions.
- **Known**: The rules of chess and the mechanics of how pieces move are perfectly known to the agent.

Crossword Puzzle

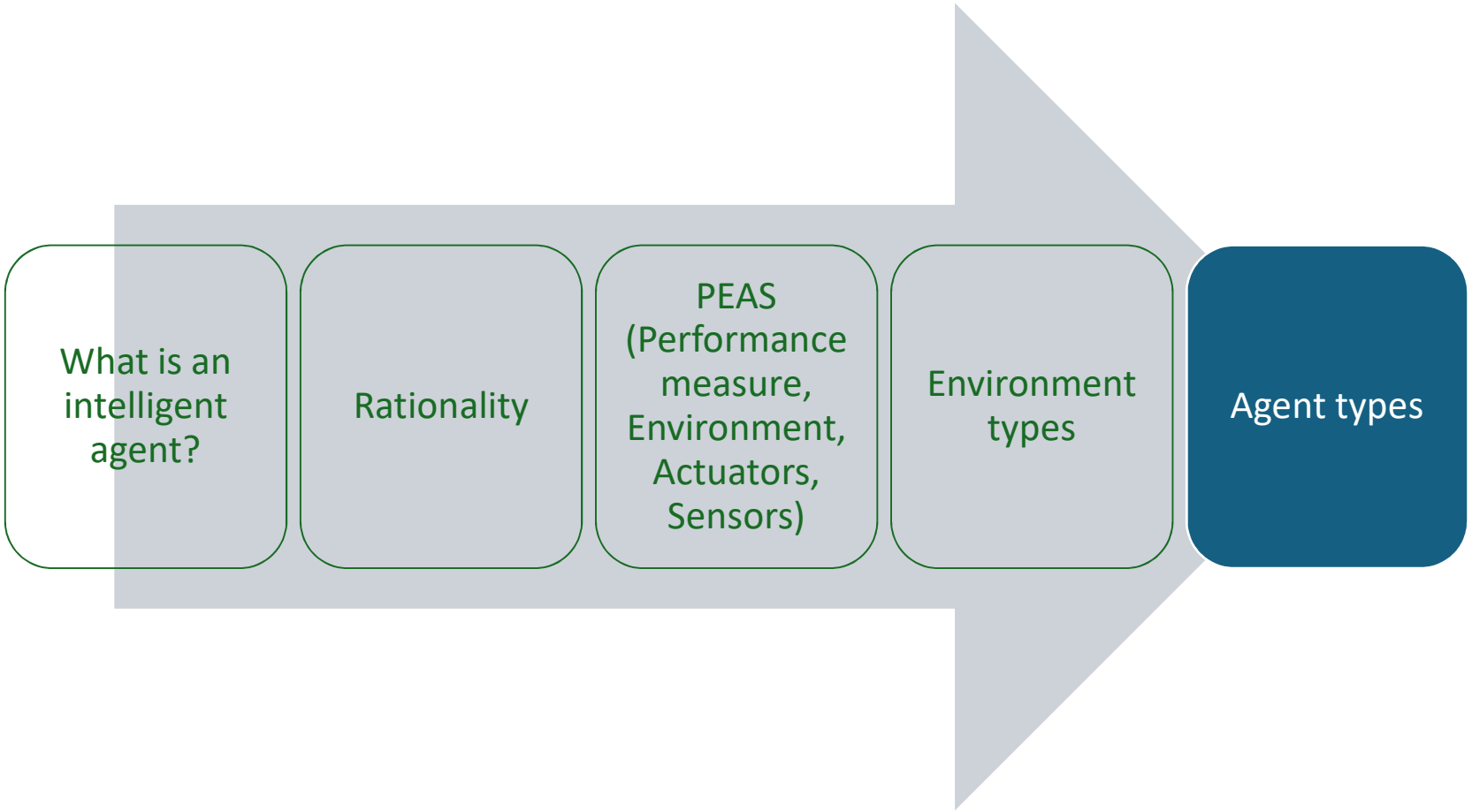
- **Fully Observable**: All filled and empty squares are entirely visible to the agent from the start.
- **Single-Agent**: The agent operates completely alone; no other entity is actively changing the board or competing.
- **Deterministic**: Entering a word has a 100% predictable outcome on the puzzle state.
- **Sequential**: Choosing a word for 1-Across constrains the letters available for 1-Down, 2-Down, and so on.
- **Static**: The puzzle grid never changes on its own while the agent takes time to calculate a word.
- **Discrete**: The grid has a finite number of squares, and the vocabulary consists of a finite set of words.
- **Known**: The entire environment's rules (how words intersect and fit) are completely defined before the task begins.

Automated Taxi Driver

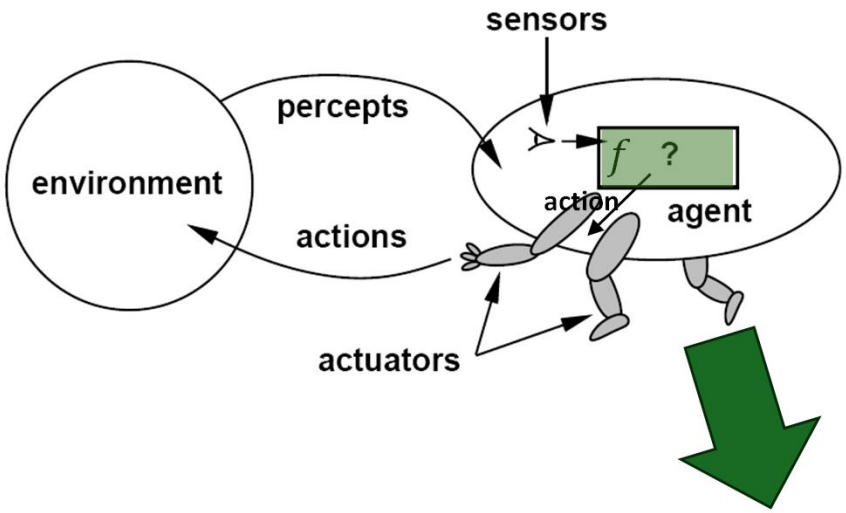
- **Partially Observable**: The agent cannot see what is around the next corner, what other drivers intend to do, or the internal condition of its own engine.
- **Multi-Agent**: The road is shared with other vehicles (agents) that can behave cooperatively or competitively.
- **Stochastic**: The environment is highly unpredictable due to random traffic patterns, sudden pedestrians, and weather changes.
- **Sequential**: A choice to take a specific highway exit dictates the driving actions required for the next several miles.
- **Dynamic**: The traffic, pedestrians, and environment constantly change while the agent's computer calculates its next move.
- **Continuous**: Variables like speed, steering angle, distance to other cars, and time are continuous numerical values.
- **Unknown**: While the rules of the road are known, the exact physical interactions (like tire traction on a patch of ice) must be learned through experience.

- According to Russell and Norvig, the **toughest task environment** is one that is:
- **Partially Observable, Multi-Agent, Stochastic, Sequential, Dynamic, Continuous, and Unknown.**

Outline: Agent Types



Designing a Rational Agent



Remember the definition of a rational agent:

*"For each possible percept sequence, a rational agent should select an **action** that **maximizes its expected performance measure**, given the evidence provided by the **percept sequence** and the **agent's built-in knowledge**."*

Agent Function

- Represents the "brain"
- Assess performance measure
- Remember percept sequence
- Built-in knowledge

Hardware + an event loop

- Read the sensors
- Ask agent function for action
- Execute action

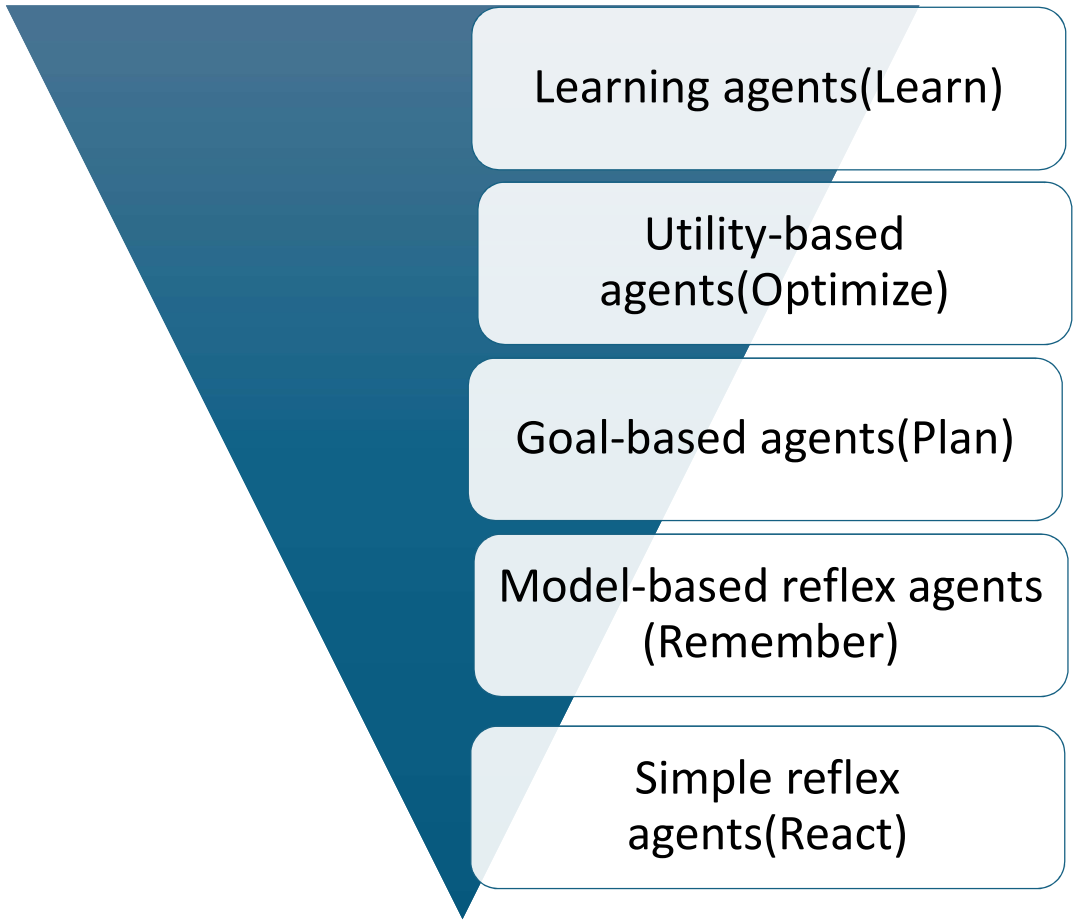
Percept to the agent function

$a = f(p)$

Action from the agent function to execute

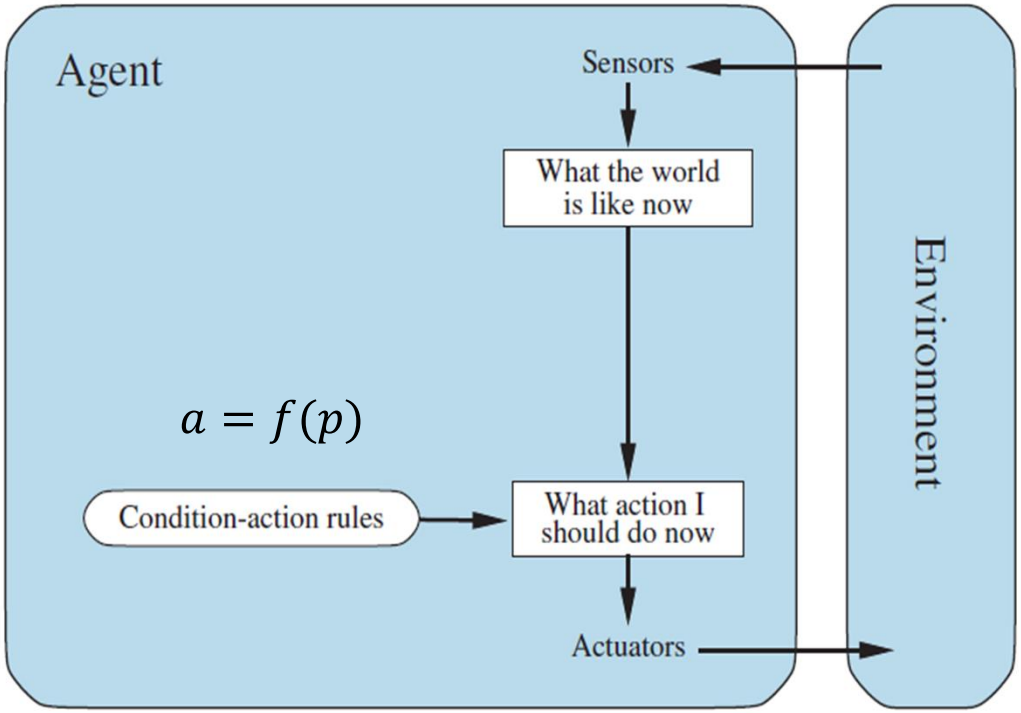
Important: Everything outside the agent function represents the environment. This includes the physical robot, its sensors and its actuators, and event loop!

Hierarchy of Agent Types based on capability



Simple Reflex Agent

- **Core Mechanism**: Action selection is based **only on the current percept**.
- **Rule Type**: Uses condition-action rules (if-then statements).
- **Limitation**: Completely ignores percept history; fails in partially observable environments.
- The **agent does not know about the performance measure**! But well-designed rules can lead to good performance.
- The agent needs no memory and ignores all past percepts.



The interaction is a sequence: $p_0, a_0, p_1, a_1, p_2, a_2, \ldots p_t, a_t, \ldots$

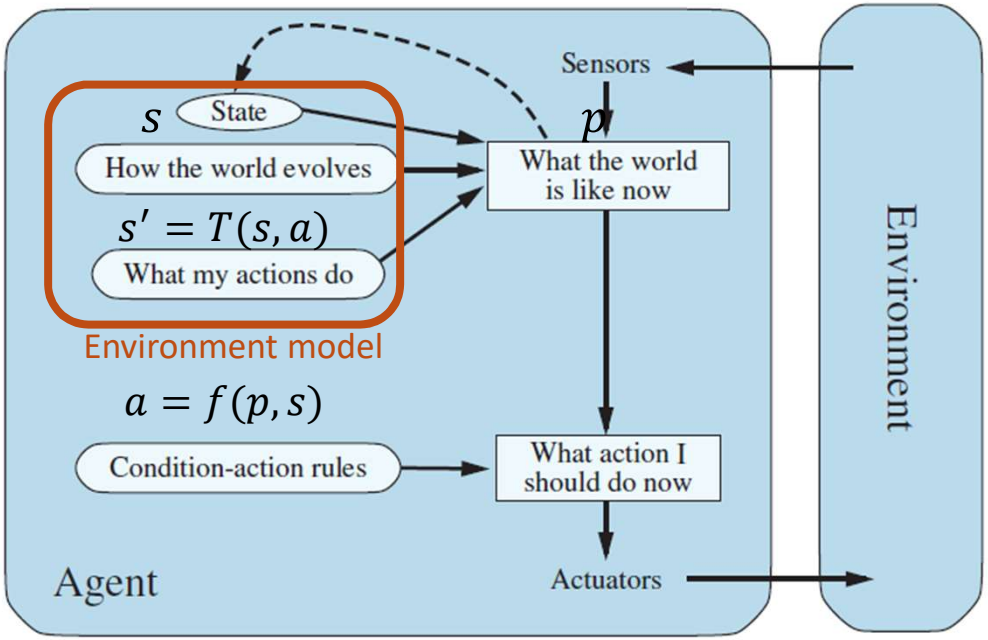
Example: A simple vacuum cleaner that uses rules based on its current sensor input. An automated thermostat. It turns on the cooling system if the current temperature exceeds 24°C.

Model-based Reflex Agent

- A **model** is the agent's internal "understanding" of how the world works.
- It is used to guess what the world looks like right now, even when the agent's sensors cannot see everything (partially observable environments).
- **What the Model Contains**
- The model consists of two specific types of background knowledge encoded into software:
  - **How the world evolves independently:** Knowledge of how the environment changes on its own over time.
  - *Example:* If a car ahead has its brake lights on, it will slow down in the next second.
  - **How the agent's own actions affect the world:** Knowledge of what happens when the agent executes a command.
  - *Example:* If the agent turns the steering wheel 90 degrees right, the car will change lanes to the right.

Model-based Reflex Agent

- **Core Mechanism: Maintains an internal state to track unobserved aspects of the current world.**
- **Rule Type: Combines current percepts with internal history to update its "model" of the world.**
- **Limitation: Can handle partial observability but still chooses actions based only on fixed rules, not goals.**
- **There is now more information for the rules to make better decisions.**



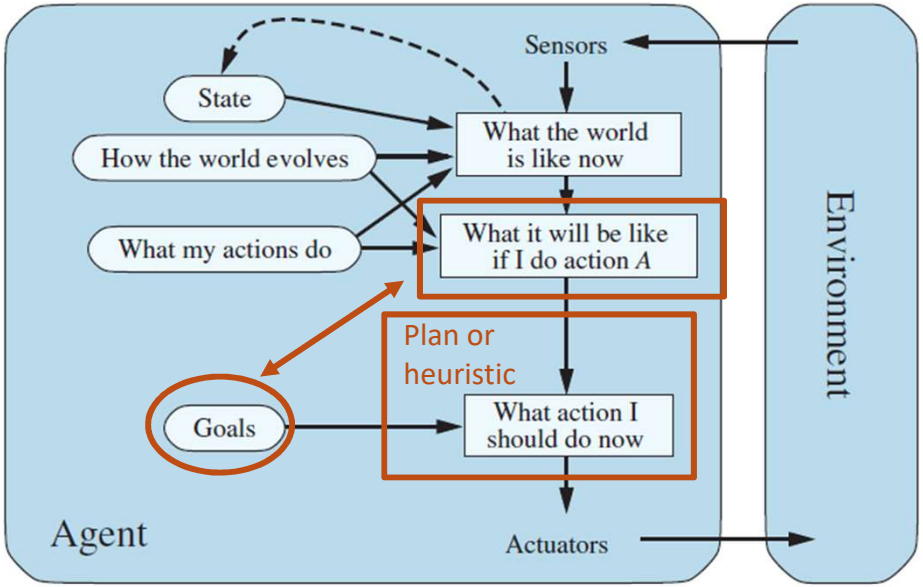
The interaction is a sequence: $p_0, s_0, a_0, p_1, s_1, a_1, p_2, s_2, a_2, p_3, \ldots, p_t, s_t, a_t, \ldots$

Example: A vacuum cleaner that remembers were it has already cleaned.

Simple Reflex	Model-Based Reflex
Uses only current percept	Uses current percept + memory
No internal state	Maintains internal state
Forgets previous observations	Remembers previous observations
Works in fully observable environments	Works in partially observable environments
No model	Has a model of the world

Goal-based Agent

- **Core Mechanism: Combines internal state tracking with explicit goal information.**
- **Rule Type: Uses search and planning algorithms to evaluate which actions will achieve the desired outcome.**
- **Limitation: Binary success metric (goal is either reached or not); cannot choose the *best* or most efficient path among multiple successful options.**



The interaction is a sequence: $\underbrace{p_0, s_0, a_0, p_1, s_1, a_1, p_2, s_2, a_2, \dots, s^{goal}}_{\text{cost}}$

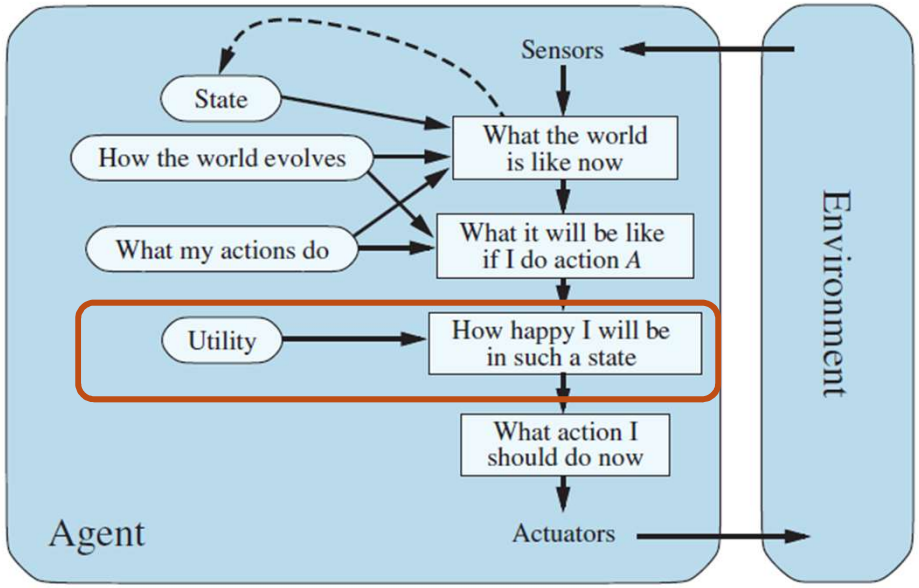
Example: Solving a puzzle. What action gets me closer to the solution?

Utility based agents

- **A simple classroom analogy :** Imagine you're choosing a seat in a classroom.
- **Goal:** Get a seat. Any empty seat satisfies the goal.
- **Utility:** You may prefer a seat near the front (better visibility), under a fan (comfort), or near the door (easy exit). You mentally assign value to these factors and choose the seat that gives you the highest overall satisfaction.
- **Utility** is simply a **numerical score that represents how desirable an outcome is**.

Utility-based Agent

- Chooses actions that maximize the desirability (utility) of the outcome, considering trade-offs such as time, cost, safety, comfort, or efficiency.
- Performance measure: The discounted sum of **expected utility over time**.



The interaction is a sequence: $\underbrace{p_0, s_0, a_0, p_1, s_1, a_1 . p_2, s_2, a_2, \ldots}_{\text{reward}}$

Example: An autonomous Mars rover prefers states where its battery is not critically low.

Learning Agents

- **Core Mechanism**: Separates the agent into a **learning element** and an **execution element** to adapt to unknown environments.
- **Four Sub-components**:
  - **Learning Element**: Makes improvements based on feedback. This is the component that actually learns. It receives feedback from the critic.
  - **Performance Element**: This is the **brain that decides what action to take**.
  - **Critic**: How well did the agent perform?
  - **Problem Generator**: Suggests new exploratory actions to discover better behaviors.

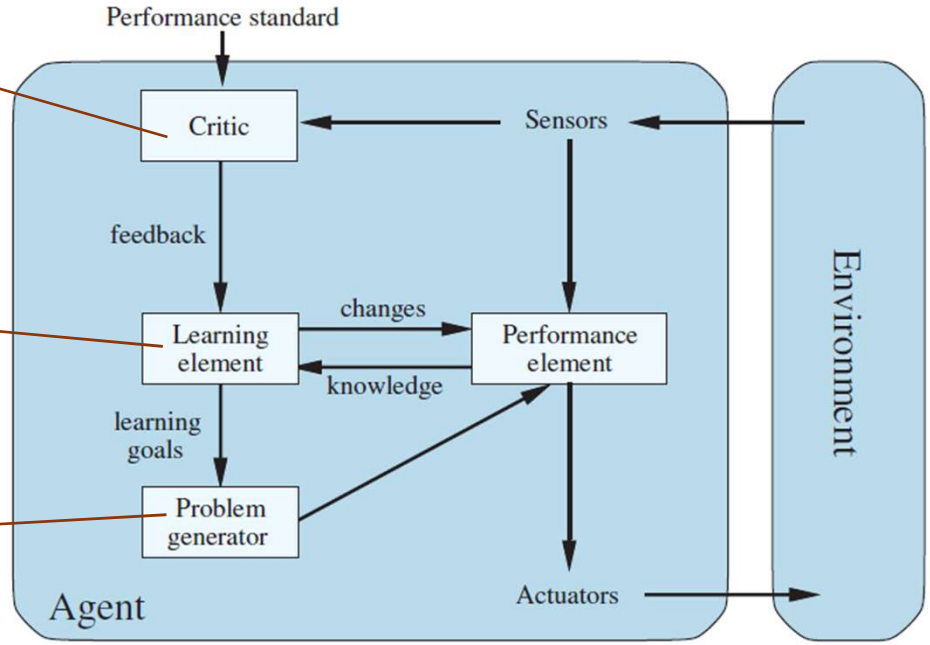
Agents that Learn

- **How can an agent improve its performance over time instead of being programmed for every possible situation?**
- Unlike the previous agents, a learning agent **becomes better with experience**.

How is the agent currently performing?

Updates how the performance element chooses actions.

Generate actions for exploration



Decision-Making Process

Agent Type	How It Decides
Simple Reflex	"What do I see now?"
Model-Based	"What do I see now, and what do I already know about the world?"
Goal-Based	"Which action will help me achieve my goal?"
Utility-Based	"Which action gives me the highest benefit or satisfaction?"
Learning	"How can I improve my future decisions using experience?"

Feature	Simple Reflex Agent	Model-Based Reflex Agent	Goal-Based Agent	Utility-Based Agent	Learning Agent
Decision Basis	Current percept only	Current percept + internal state	Current percept + model + goal	Current percept + model + goal + utility	Learns from experience and feedback
Memory (Internal State)	✗ No	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Uses a Model of the Environment	✗ No	✓ Yes	✓ Yes	✓ Yes	✓ Yes (can improve it over time)
Uses Goals	✗ No	✗ No	✓ Yes	✓ Yes	✓ Yes (if applicable)
Uses Utility Function	✗ No	✗ No	✗ No	✓ Yes	✓ Yes (if utility-based)

Learns from Experience	✗ No	✗ No	✗ No	✗ No	☑ Yes
Considers Future Consequences	✗ No	Limited	☑ Yes	☑ Yes	☑ Yes
Chooses Between Multiple Good Solutions	✗ No	✗ No	✗ No (any goal-achieving solution is acceptable)	☑ Yes (chooses the best one)	☑ Yes (improves choices over time)
Works in Partially Observable Environment	✗ Poorly	☑ Yes	☑ Yes	☑ Yes	☑ Yes

Adapts to New Situations	✗ No	✗ No	✗ No	✗ No	☑ Yes
Computational Complexity	Very Low	Low	Moderate	High	Highest
Typical Applications	**Automatic sliding door, Automatic hand dryer, Motion-sensor light, Thermostat**	**Robot vacuum cleaner, Warehouse robot, Maze-solving robot, Line-following robot with obstacle memory**	**GPS route planner, Chess-playing program, 8-puzzle solver, Robot path planner**	**Investment advisor, Medical treatment planner, Airline route planner, Ride-sharing route optimizer**	**ChatGPT, Self-driving car, Spam email filter, Netflix/YouTube recommendation system, AlphaGo, Face recognition system, Speech recognition system**